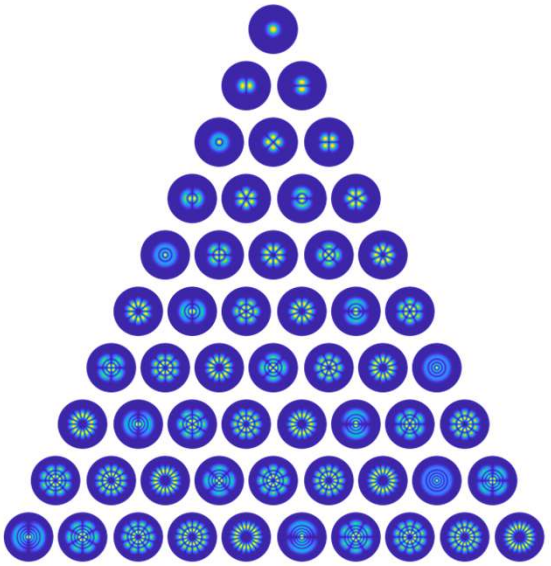
University of Stuttgart
Institute of electrical and optical communications

Advanced optical fiber communications

Wintersemester 2025/26

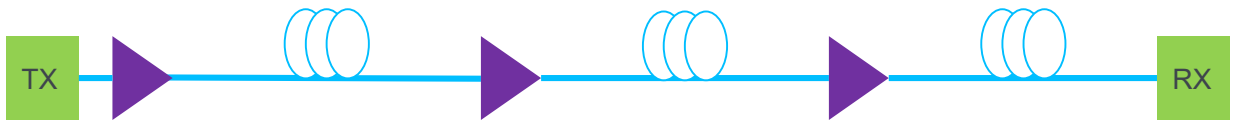
Lecture 12

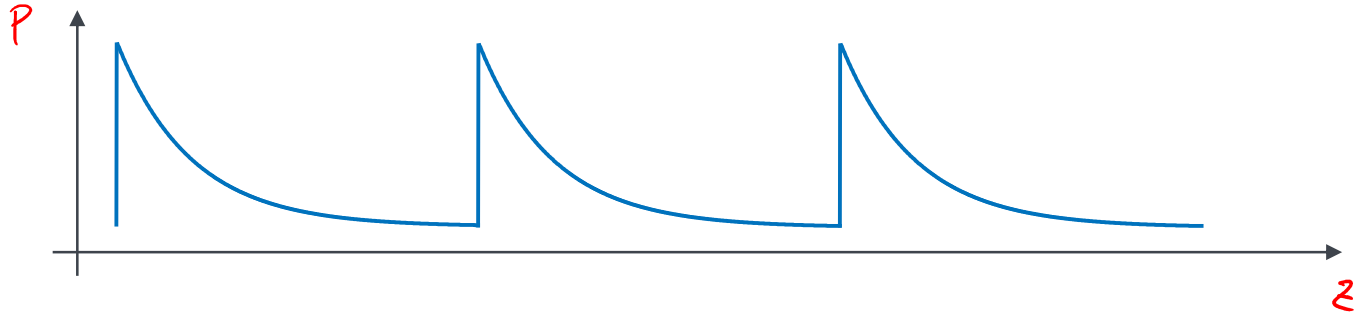


Prof. Dr.-Ing. Georg Rademacher

Power profile in amplified transmission link

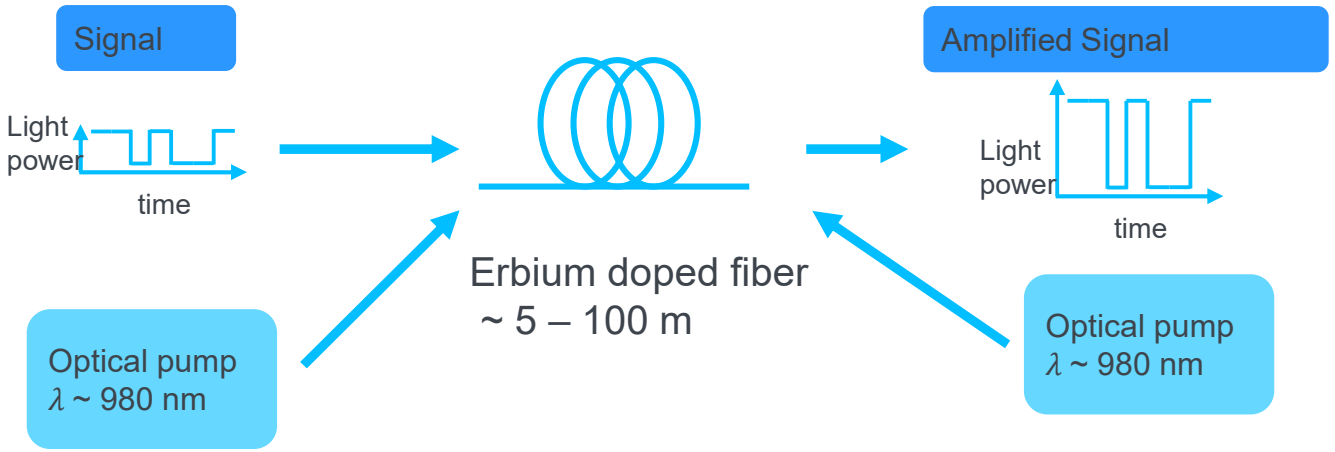
University of Stuttgart
Germany

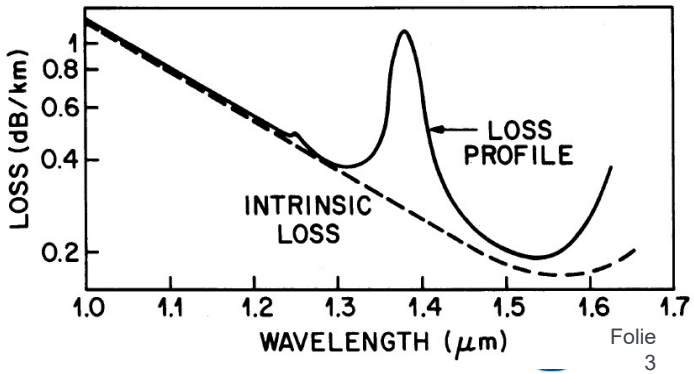




Erbium doped fiber amplifier (EDFA)

University of Stuttgart
Germany



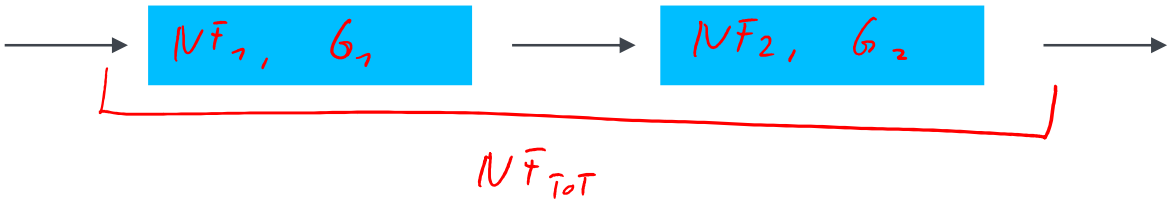


Noise in long distance transmission

University of Stuttgart
Germany

$$NF = 2 n_{sp} \left(1 - \frac{1}{G}\right) + \frac{1}{G} = \frac{1}{G}\left(1 + 2 n_{sp} (G - 1)\right) \approx 2 \cdot n_{sp} \quad (G >> 1)$$

$\Rightarrow$ minimum $NF = 3\,dB$



$$NF_{TOT} = NF_1 + \frac{NF_2 - 1}{G_1}$$

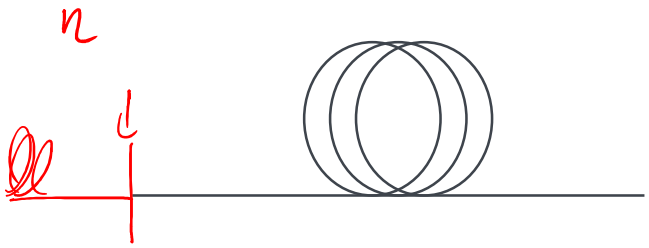
Noise in long distance transmission

University of Stuttgart
Germany

$$NF_1 = \frac{1}{G_1} = \frac{1}{\eta}$$

$$NF_2 = 2n_{sp}\left(1 - \frac{1}{G_2}\right) + \frac{1}{G_2}$$

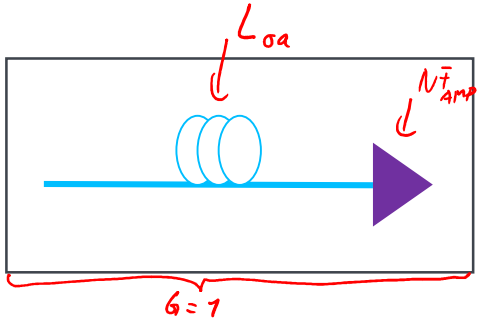
$$NF_{TOT} = \frac{1}{\eta} + \frac{NF_2 - 1}{\eta} = \frac{NF_2}{\eta}$$



$\rightarrow \triangleright NF_2$

Noise in long distance transmission

University of Stuttgart
Germany



$$N\bar{F}_{seg} = G + G(N\bar{F}_{amp} - 1) = G \cdot N\bar{F}_{AMP}$$

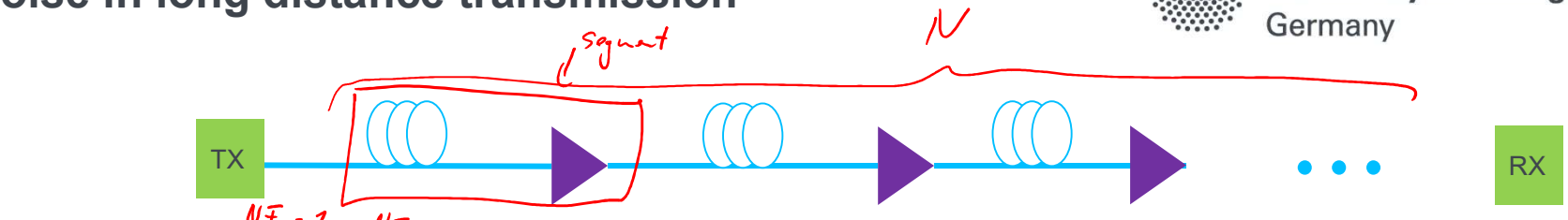
$$N\bar{F}_{seg} = 1 + 2n_{sp}(G-1)$$

Loss / attenuation : $\exp(-2\alpha L_{oa})$ $\rightarrow$ $G = \exp(2 \cdot \alpha \cdot L_{oa})$

α: Field-attenuation

Noise in long distance transmission

University of Stuttgart
Germany



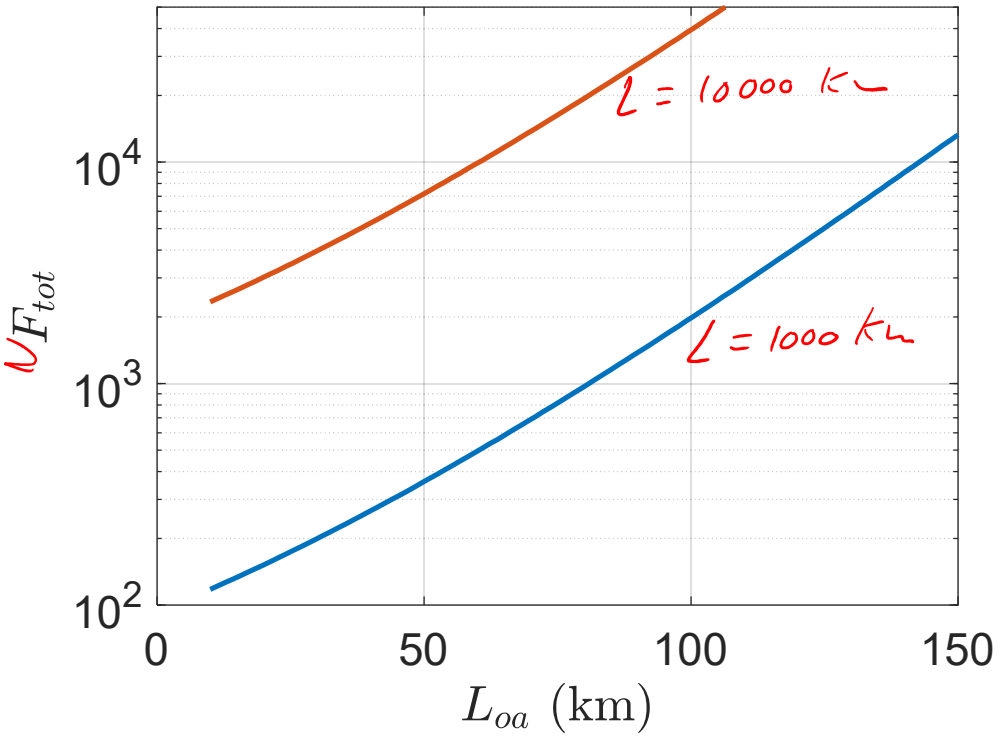
$$N\bar{F}_{TOT} = \bar{F}_1 + \frac{N\bar{F}_2 - 1}{G_1} + \frac{N\bar{F}_3 - 1}{G_1 \cdot G_2} + \dots$$

$$N\bar{F}_{TOT} = 1 + N(N\bar{F}_{seg} - 1) = 1 + 2N n_{sp}(G - 1)$$

Link distance: $L \rightarrow N = \frac{L}{L_{oa}}$, $G = \exp(2 \cdot \alpha \cdot L_{oa})$

$$\rightarrow N\bar{F}_{TOT} = 1 + 2 n_{sp} \frac{L}{L_{oa}} (\exp(2 \alpha L_{oa}) - 1)$$

Noise in long distance transmission



Optical Signal to Noise Ratio (OSNR)

University of Stuttgart
Germany

$$OSNR = \frac{P_0}{NF \cdot h \cdot f \cdot \Delta f}$$

$$\Delta f = 12{,}5\,GHz \hat{=} 0{,}1\,nm \quad (\sim 1550\,nm)$$

$$OSNR_{dB} = 10 \cdot \log_{10}(OSNR)$$

$$OSNR_{dB} = 58\,dB + P_0|_{dBm} - NF|_{dB}$$

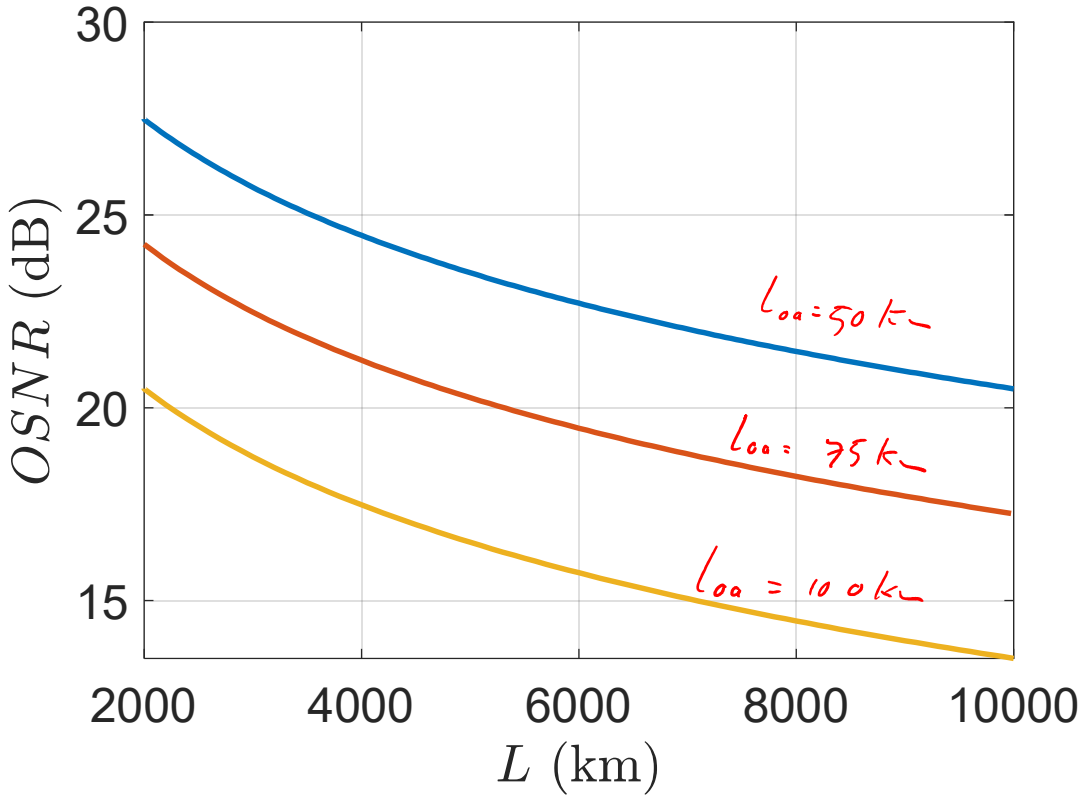
OSNR in long distance transmission

University of Stuttgart
Germany

$$OSNR|_{dB} = 58\,dB + P_0|_{dBm} - NF_{AMP} - \alpha \cdot L_{oa}|_{dB} - 10 \cdot \log(N)|_{dB}$$

OSNR in long distance transmission

University of Stuttgart
Germany



$\alpha = 0{,}2$ dB/km

$NF = 4{,}5$ dB



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Germany

Concept Test

Concept test

University of Stuttgart
Germany

What is the OSNR for P_0 = 2 mW, F_{Amp} = 4 dB, α = 0.2 dB/km, L_{oa} = 80 km, and N = 8?

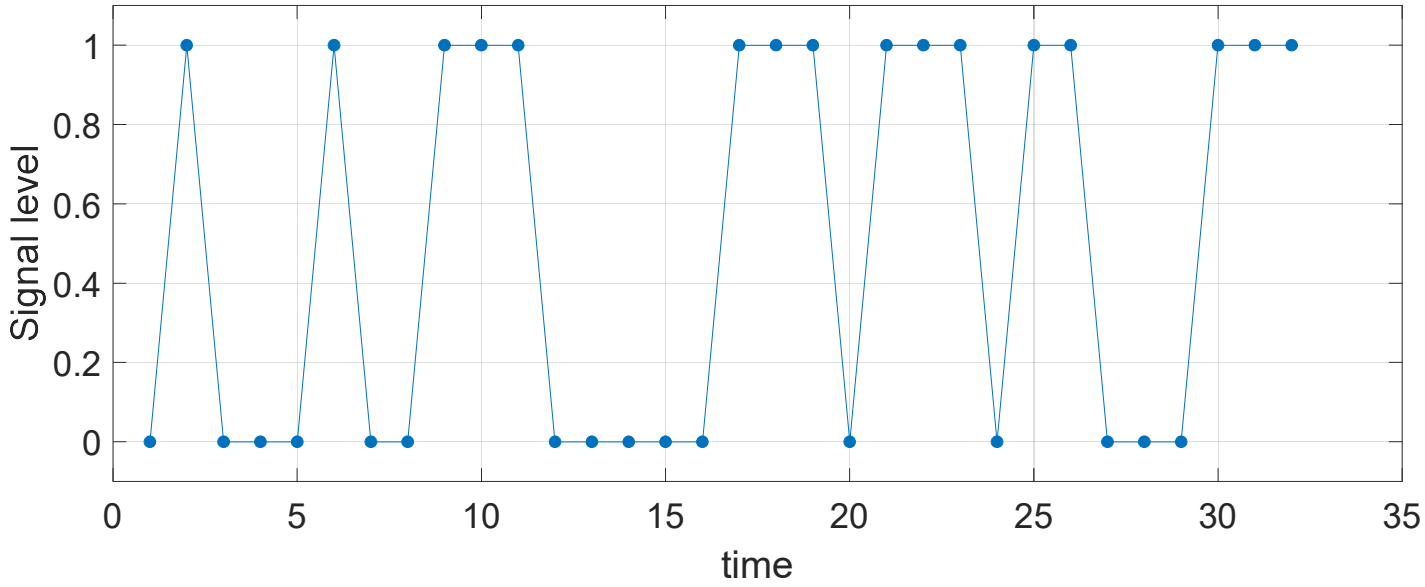
- **A** 44 dB
- **B** 30 dB
- **C** 50 dB

58 dB + 3 dBm - 4 dB - 16 dB -

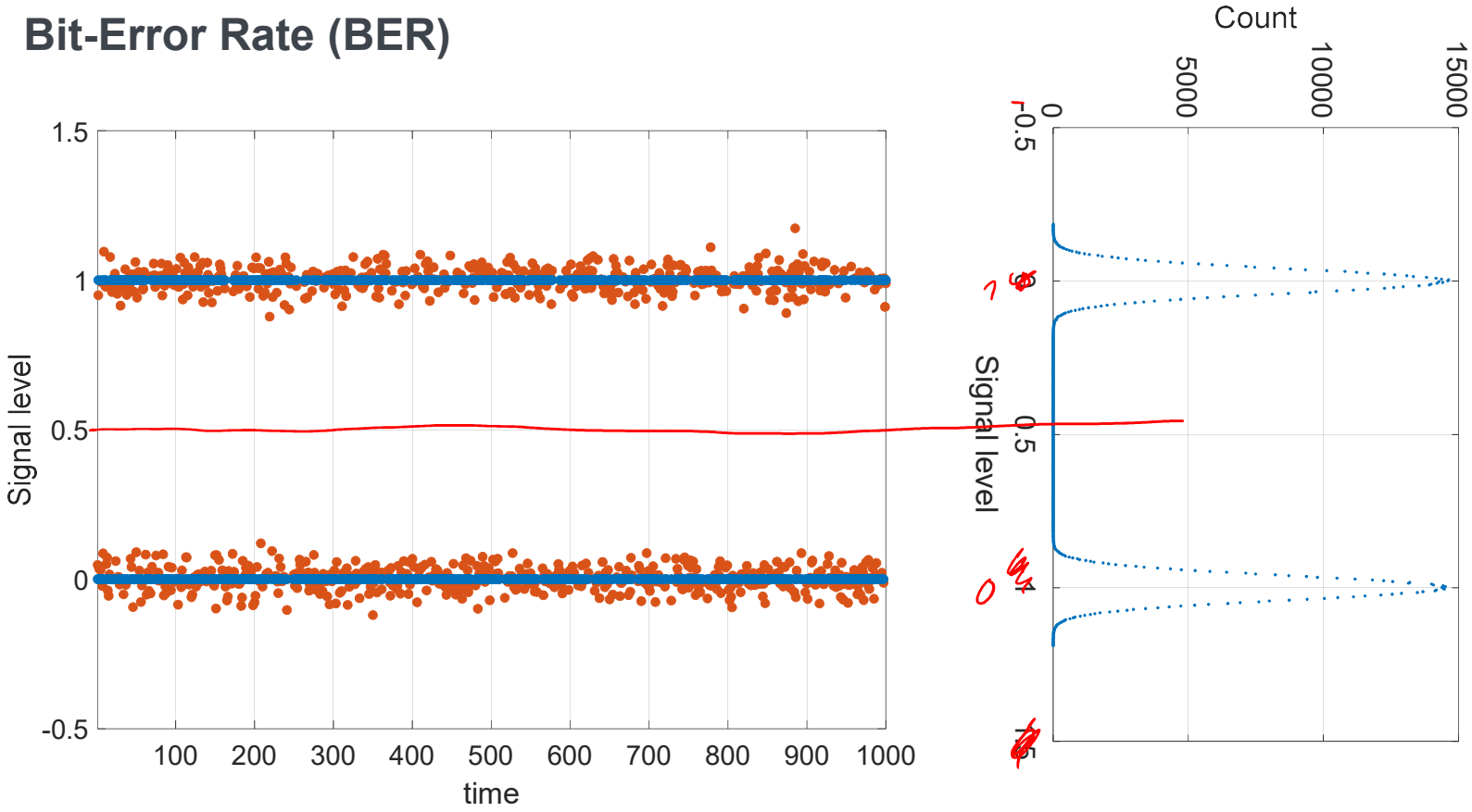
https://ilias3.uni-stuttgart.de/vote/KPS5

Bit-Error Rate (BER)

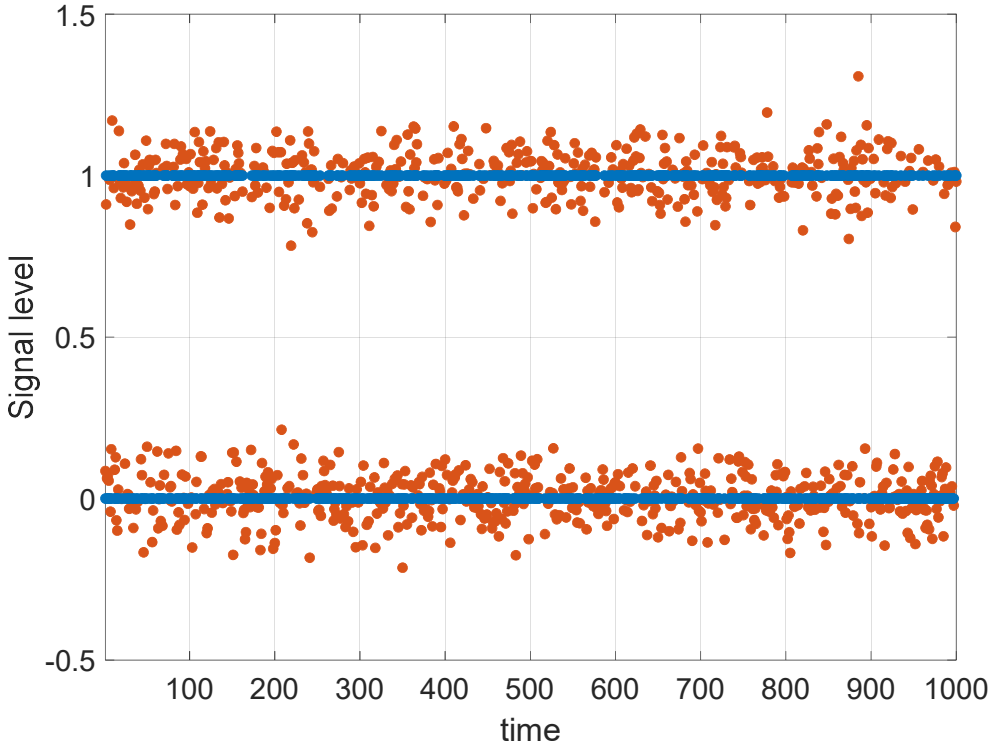
University of Stuttgart
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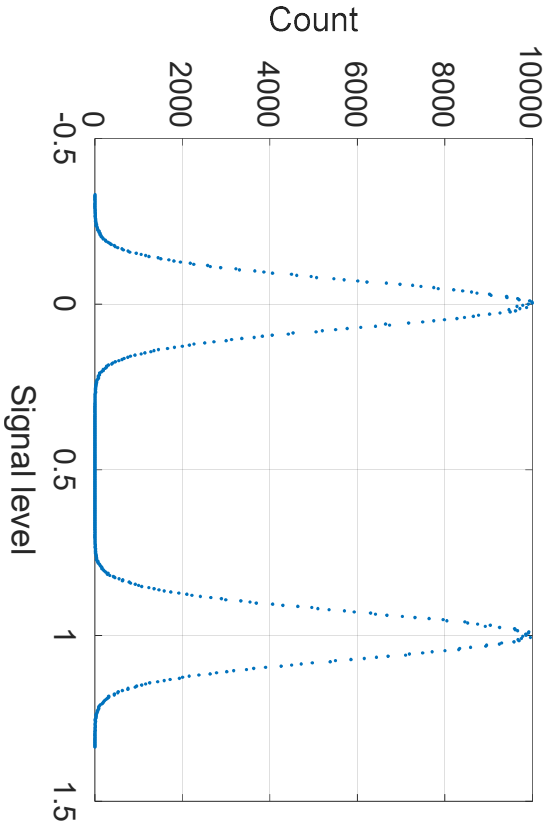


Bit-Error Rate (BER)

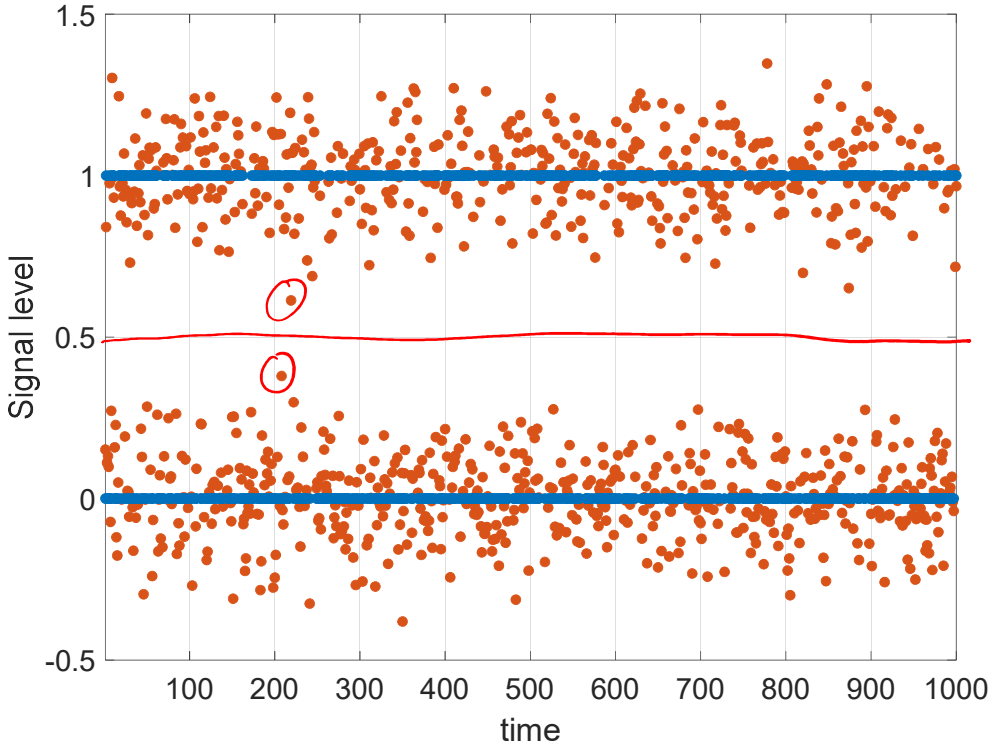


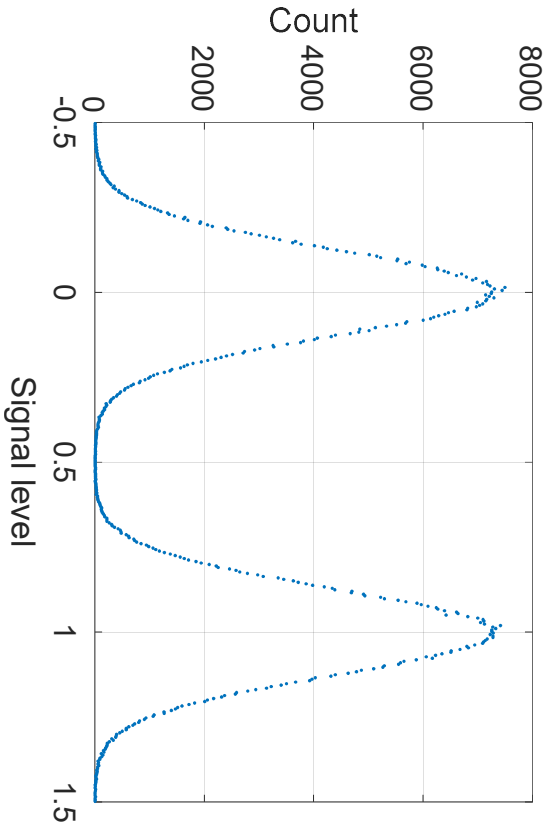
Bit-Error Rate (BER)



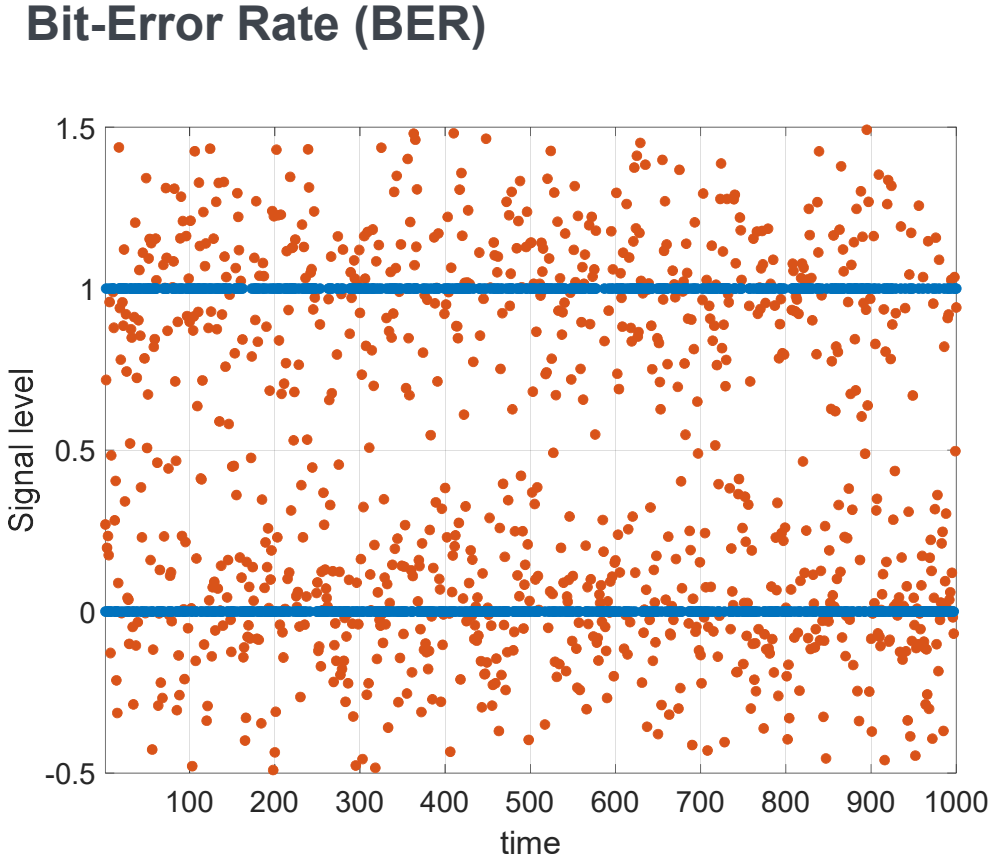


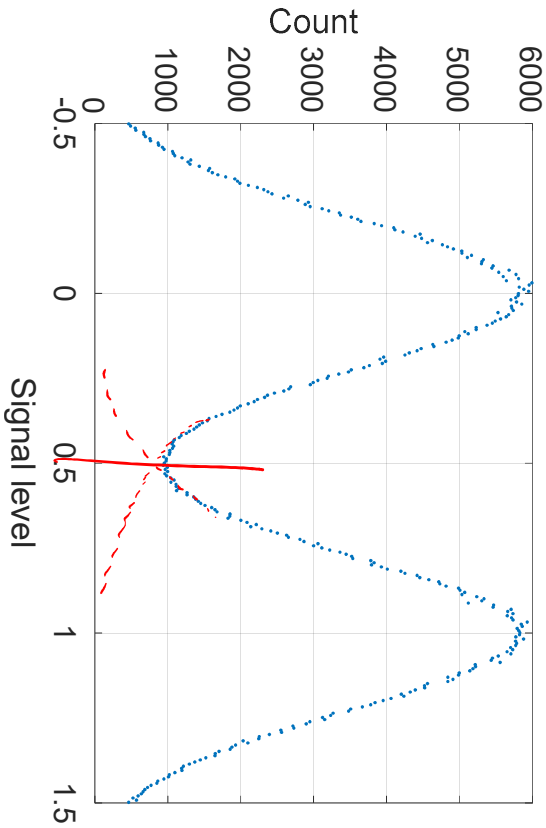
Bit-Error Rate (BER)





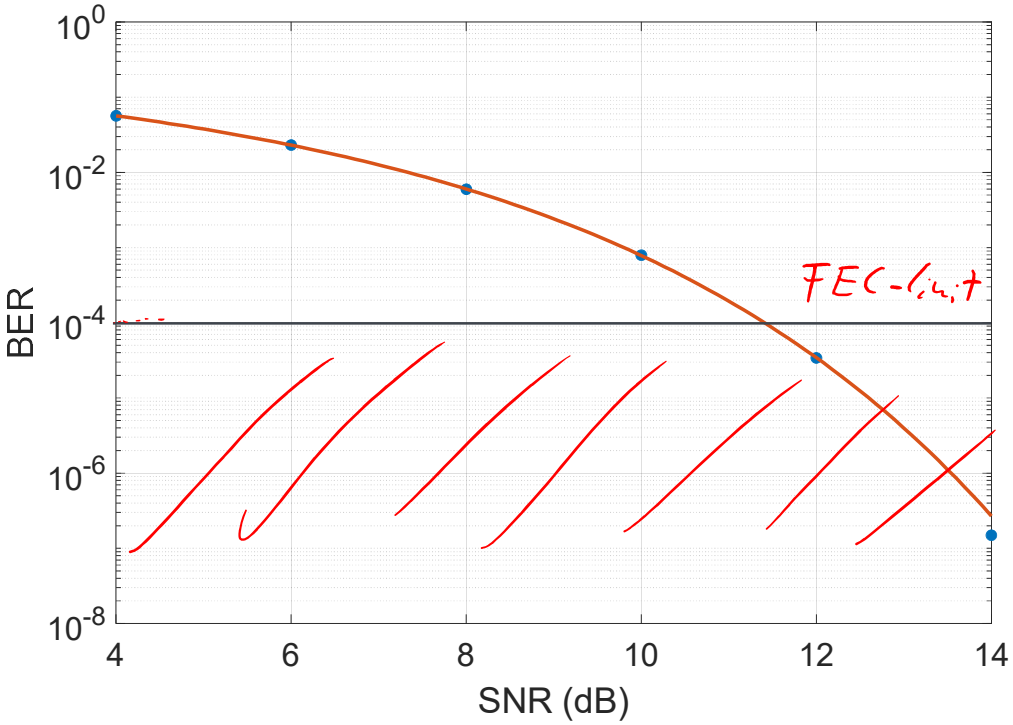
Bit-Error Rate (BER)





BER vs SNR for OOK

University of Stuttgart
Germany



$$BER = \frac{\#\ \text{wrong bits}}{\text{total}\ \#\ \text{bits}}$$

$$BER_{TH} = 0{,}5\ \text{erfc}\left(\sqrt{\frac{SNR}{2}}\right)$$

$$\text{erfc} = 1 - \text{erf}$$

$$\text{erf} = \frac{2}{\sqrt{\pi}} \int_0^{+} \exp(-\tau^2)\, d\tau$$

Forward Error Correction (FEC)

University of Stuttgart
Germany

(3,1) repitition code

Payload bit	Msg	received Msg.	interpreted Msg
0	000	000	0
		001	0
		010	0
		100	0
1	111	111	1
		110	1
		101	1
		011	1

Forward Error Correction (FEC)



University of Stuttgart
Germany

Information-Theoretic Tools for Optical Communications Engineers

(Invited Tutorial)

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TABLE I
HARD-DECISION FEC THRESHOLDS OF CERTAIN CODES RECOMMENDED FOR OPTICAL COMMUNICATIONS

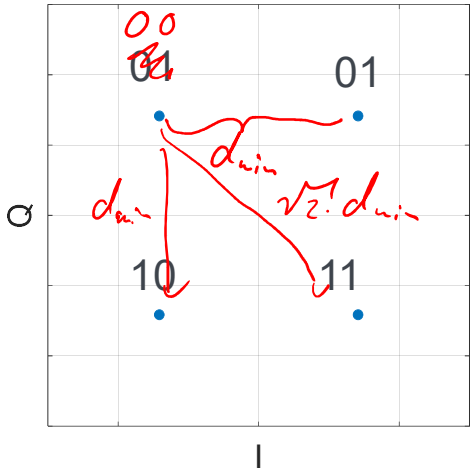
OH	Code type	Source	post-FEC BER	p
2.7%	Reed–Solomon (RS) (528,514) "KR4"	[26][a]	10^{-15}	$2.18 \cdot 10^{-5}$
5.8%	RS(544,514) "KP4"	[26][a]	10^{-15}	$2.26 \cdot 10^{-4}$
6.7%	RS(255,239)	[27][a]	10^{-15}	$8.27 \cdot 10^{-5}$
6.7%	Bose–Chaudhuri–Hocquenghem (BCH)+BCH	[28, App. I.3]	10^{-15}	$3.15 \cdot 10^{-3}$

Coherent Signals

4 QAM / QPSK

$\log_2(M) = 2$ bit/symbol

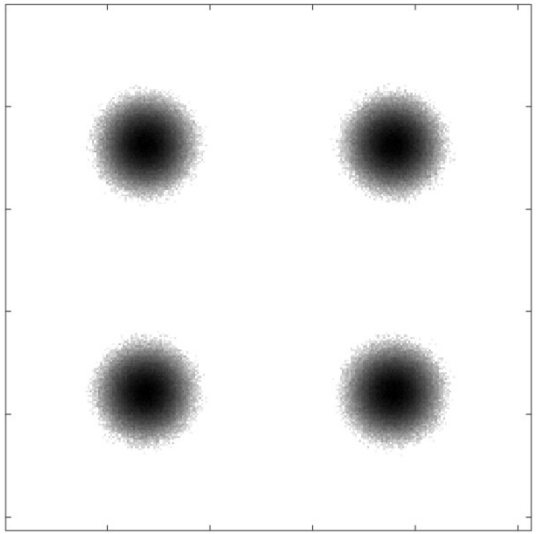
Gray - Coding



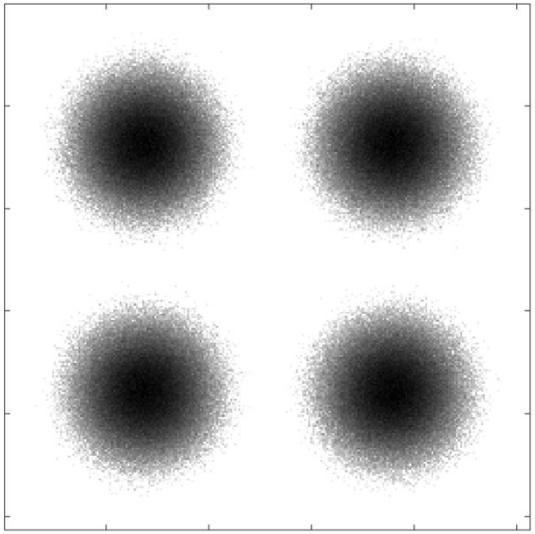
Coherent Signals

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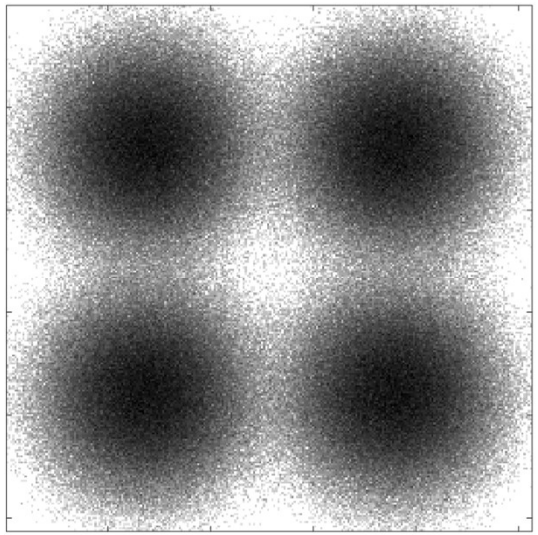
20 dB SNR



15 dB SNR



10 dB SNR



Coherent Signals

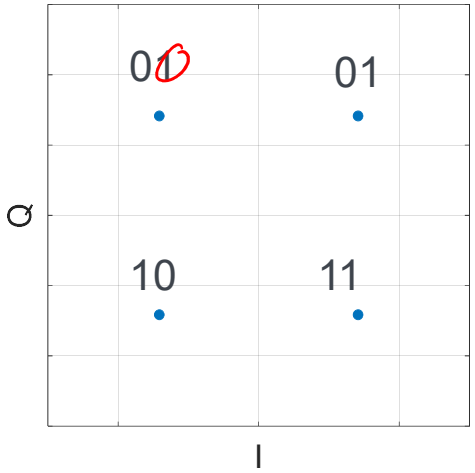
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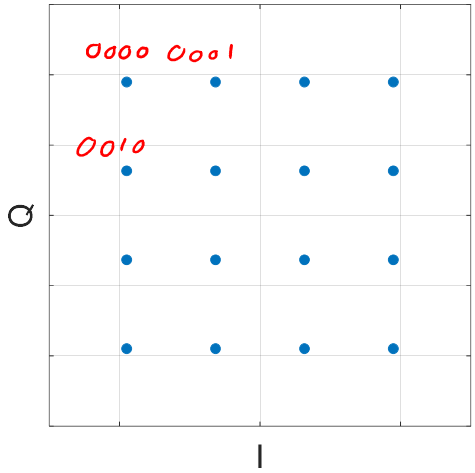
16 QAM

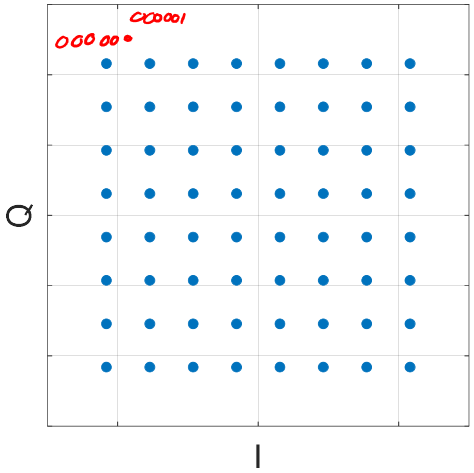
$\log_2(16) = 4\ bit/sy-b$

64 QAM

$\log_2(64) = 6\ bit/sy-b$







Data-rate $25\ GBd \cdot n\ \frac{bit}{sy-b} \cdot \frac{1}{(1 + OH_{FEC})}$



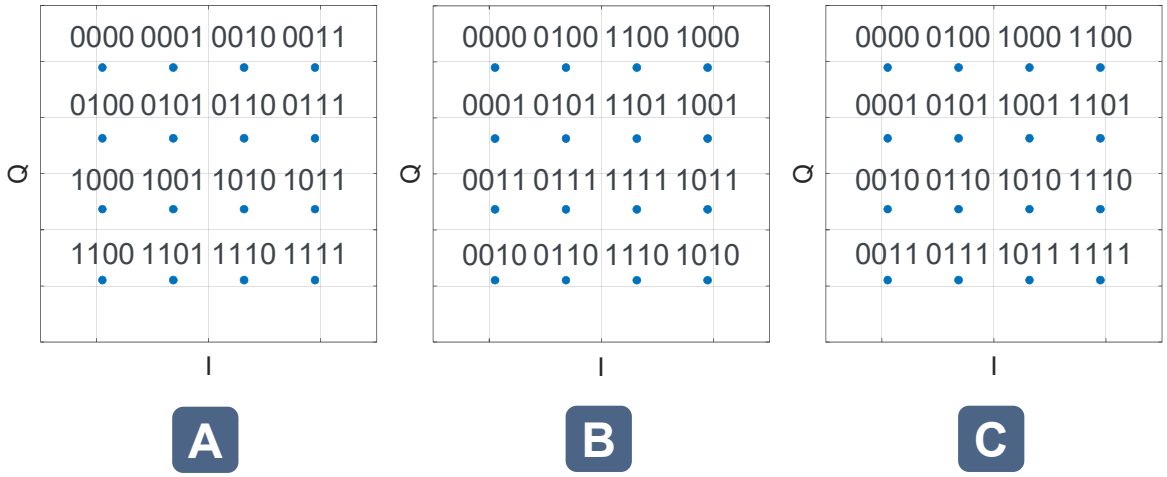
University of Stuttgart
Germany

Concept Test

Concept test

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Germany

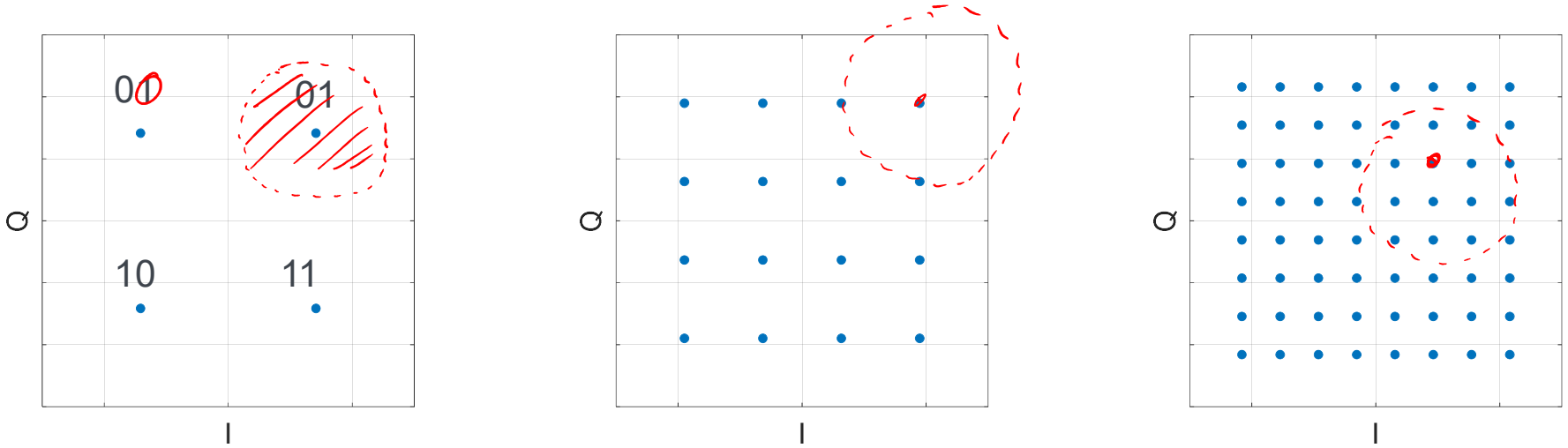
Which of the shown constellation diagrams is Gray encoded?



https://ilias3.uni-stuttgart.de/vote/PLKM

Coherent Signals

University of Stuttgart
Germany



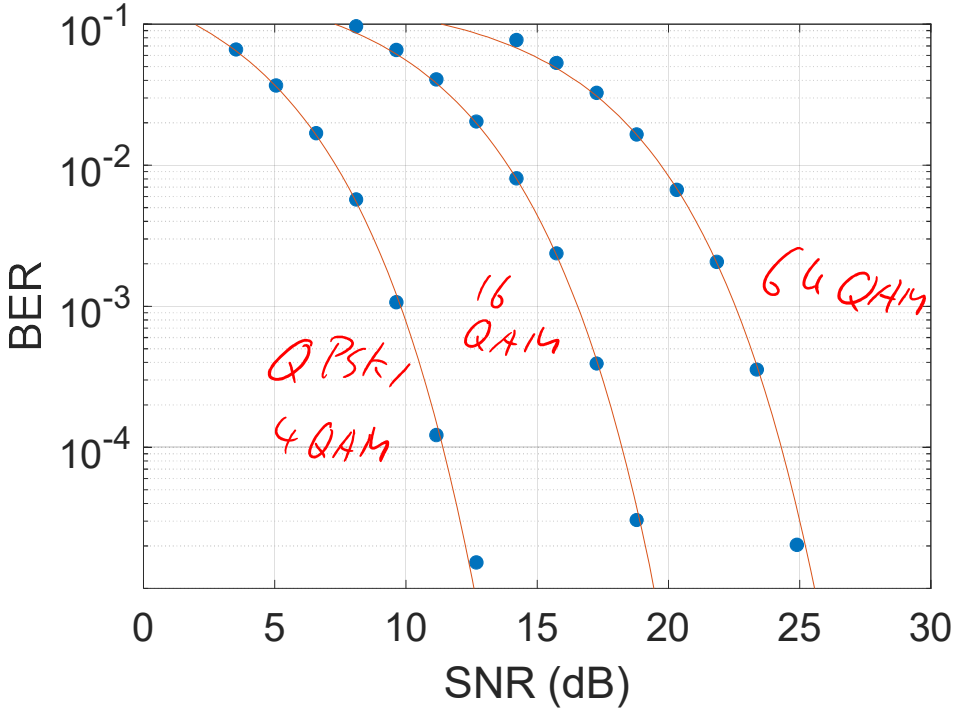
BER vs. SNR

University of Stuttgart
Germany

$$P_M = \left(1 - \frac{1}{\sqrt{M}}\right) \cdot \operatorname{erfc}\left(\sqrt{\frac{3}{2(M-1)} \cdot SNR}\right)$$

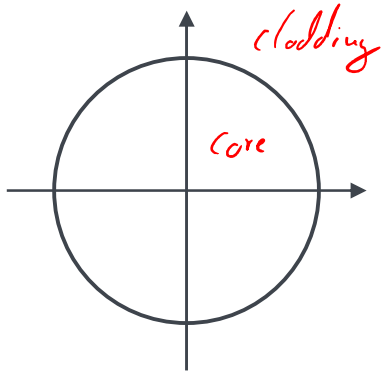
$$P_S = 1 - (1 - P_M)^2$$

$$P_e = \frac{P_S}{\log_2(M)} = BER$$



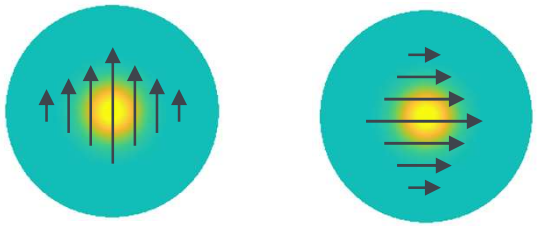
Polarization multiplex (Pol Mux)

University of Stuttgart
Germany

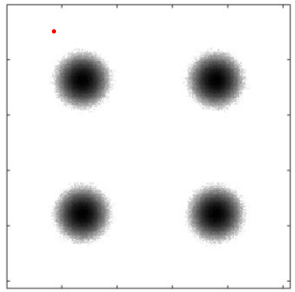


$E_y = 0$, $E_x = \psi(r,\varphi) \exp(-j\beta z)$

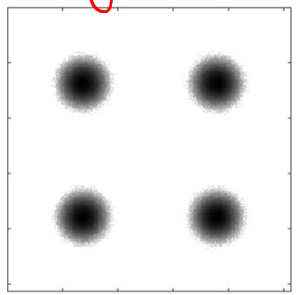
$E_y = \psi(r,\varphi) \cdot \exp(-j\beta z)$, $E_x = 0$



x- Pol

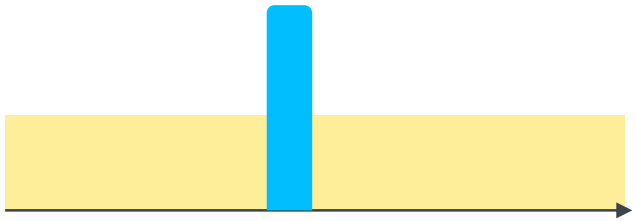


y- Pol



$25\,\text{GBd} \cdot n \frac{\text{bit}}{\text{Symb}} \cdot \frac{1}{1+OH_{FEC}} \cdot 2\,\text{pol}$

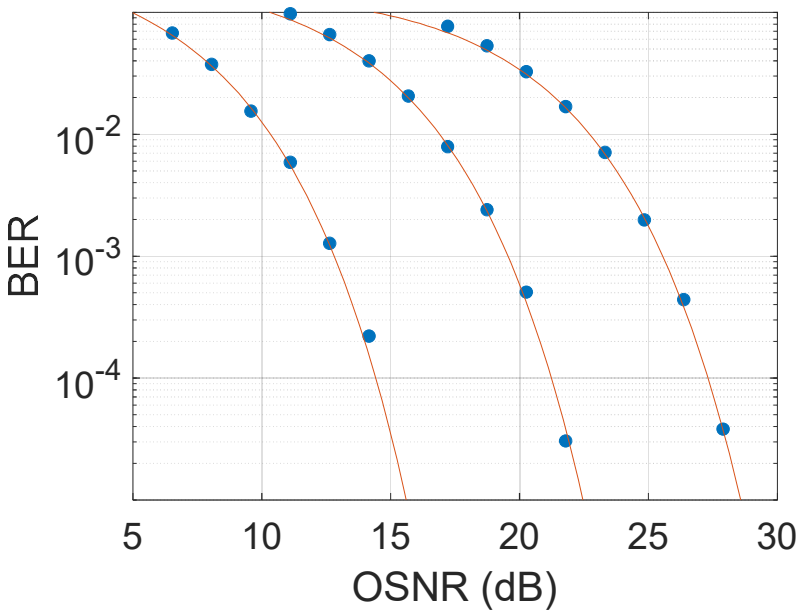
SNR vs. OSNR

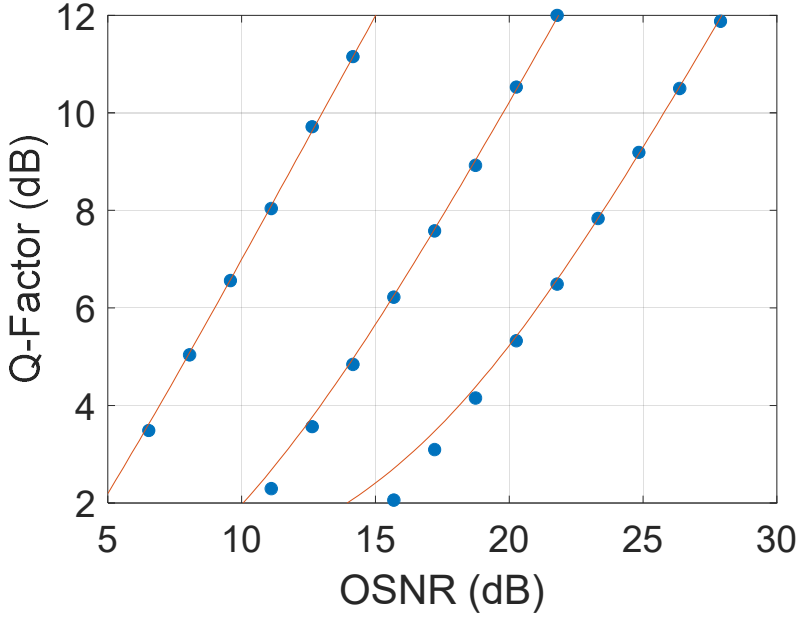




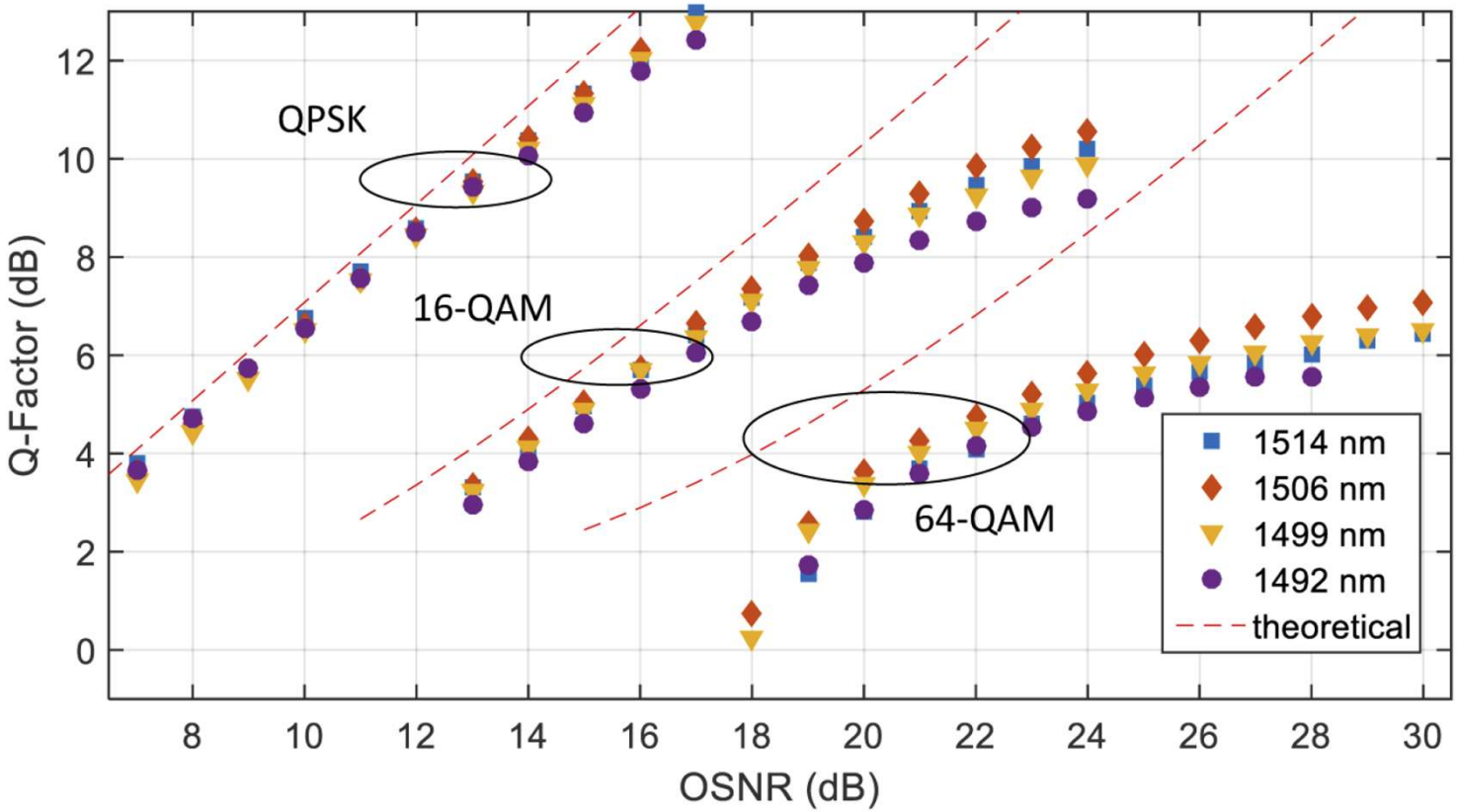
Coherent Signals

University of Stuttgart
Germany



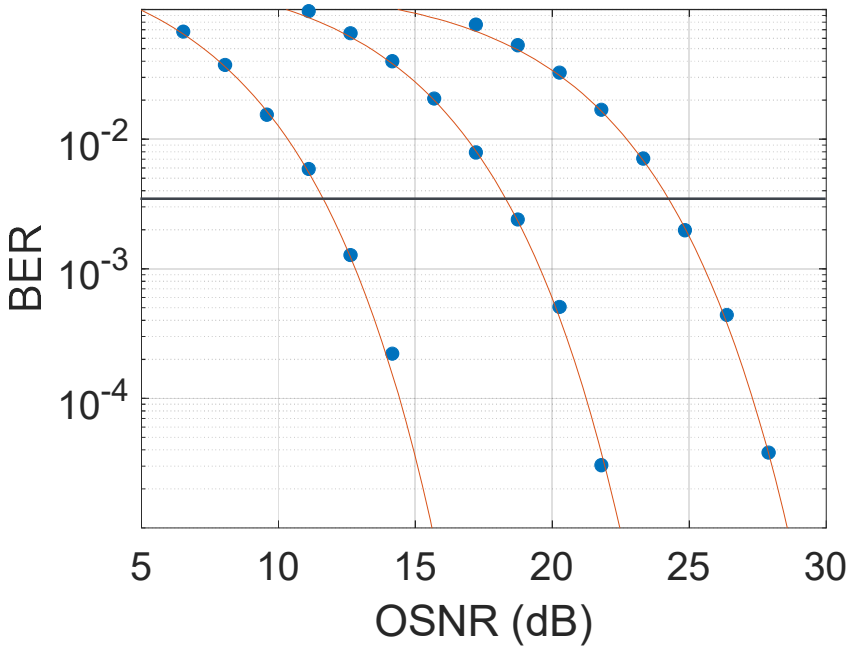


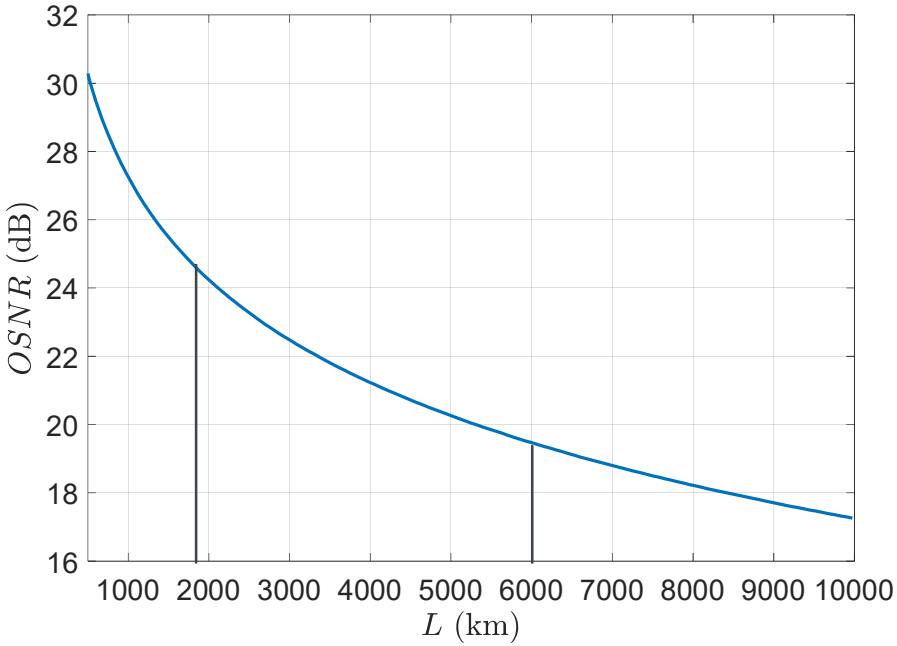
Measured performance



Long distance transmission

University of Stuttgart
Germany







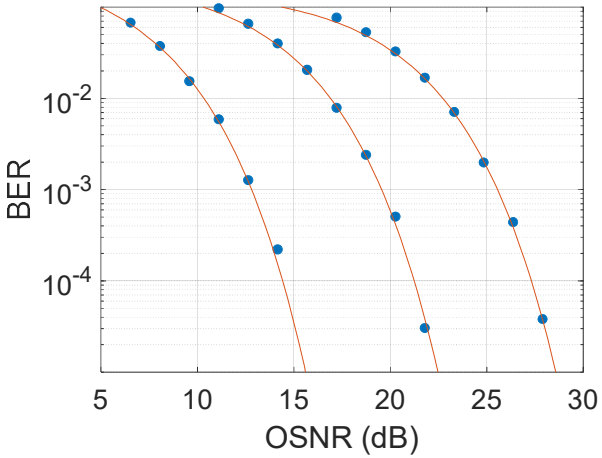
University of Stuttgart
Germany

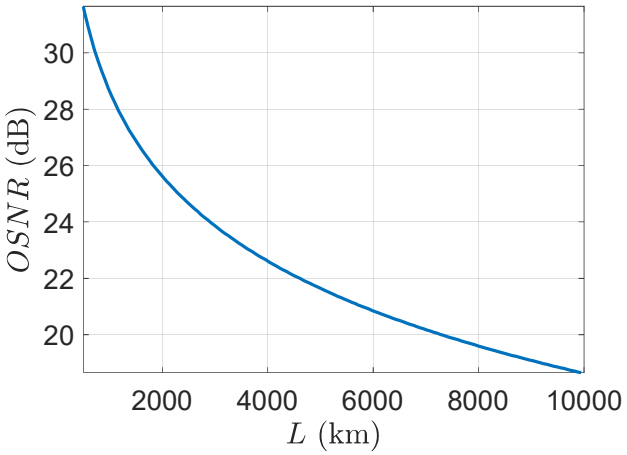
Concept Test

Concept test

University of Stuttgart
Germany

What is the maximum data rate for an FEC limit of $3 \cdot 10^{-4}$ with an overhead of 10% for L = 4500 km at 25 Gbd?

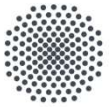




A 45.45 Gbit/s **B** 90.91 Gbit/s **C** 181.82 Gbit/s

https://ilias3.uni-stuttgart.de/vote/DC90

Pulse propagation in SMF



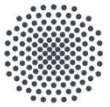
University of Stuttgart
Germany

Pulse propagation in SMF



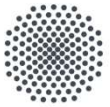
University of Stuttgart
Germany

Pulse propagation in SMF



University of Stuttgart
Germany

Pulse propagation in SMF



University of Stuttgart
Germany

Pulse propagation in SMF

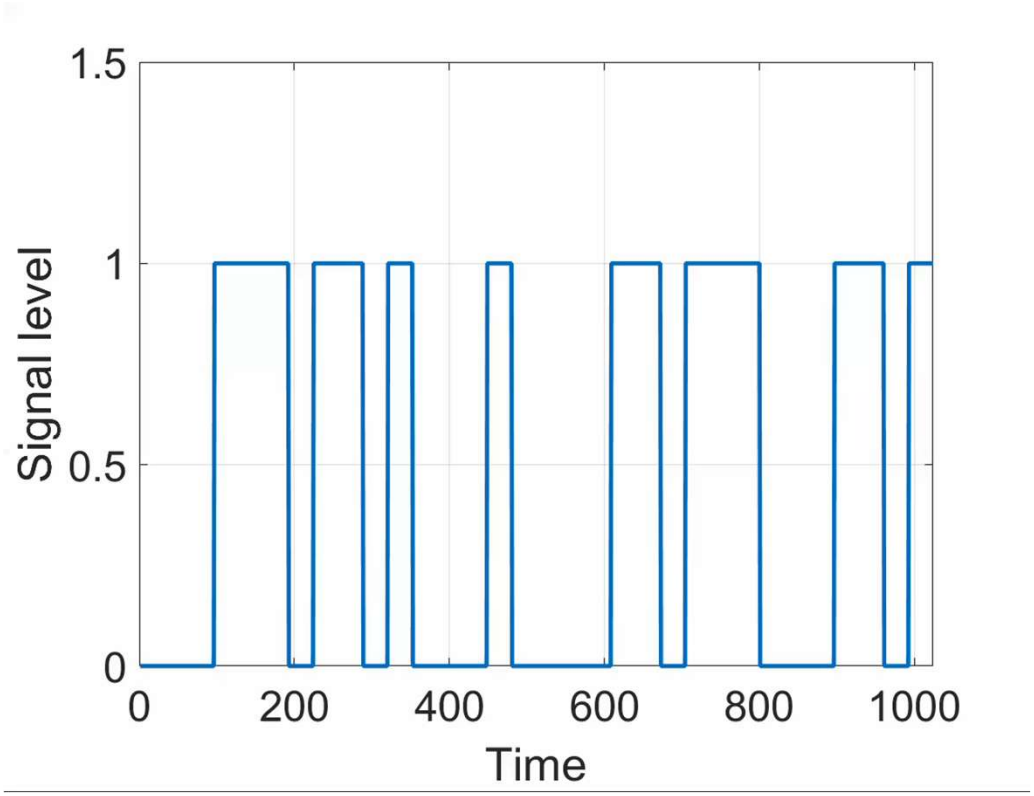


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Germany

Pulse propagation in SMF



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Germany


1.5
1
0.5
0
Signal level
0
200
400
600
800
1000
Time

University of Stuttgart
Institute of electrical and optical communications

Thank you!



Prof. Dr.-Ing. Georg Rademacher

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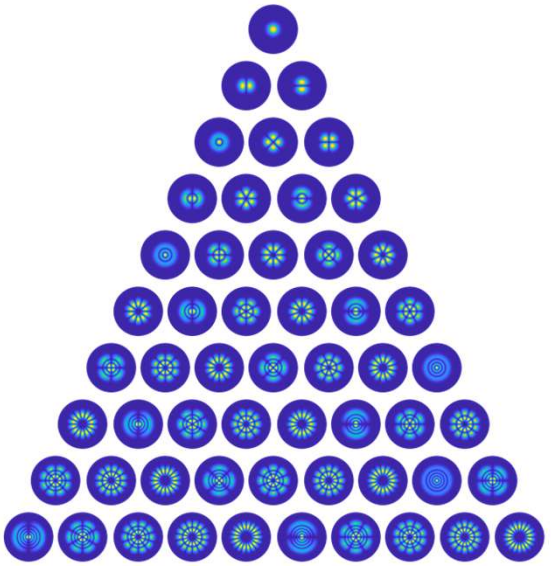


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Advanced optical fiber communications

Wintersemester 2025/26

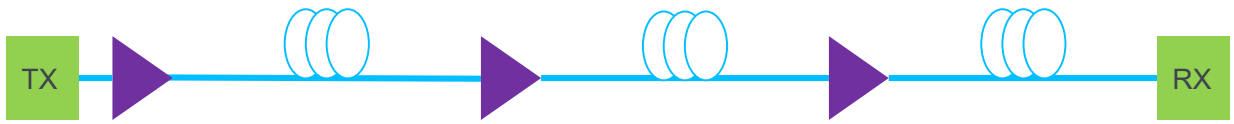
Lecture 12

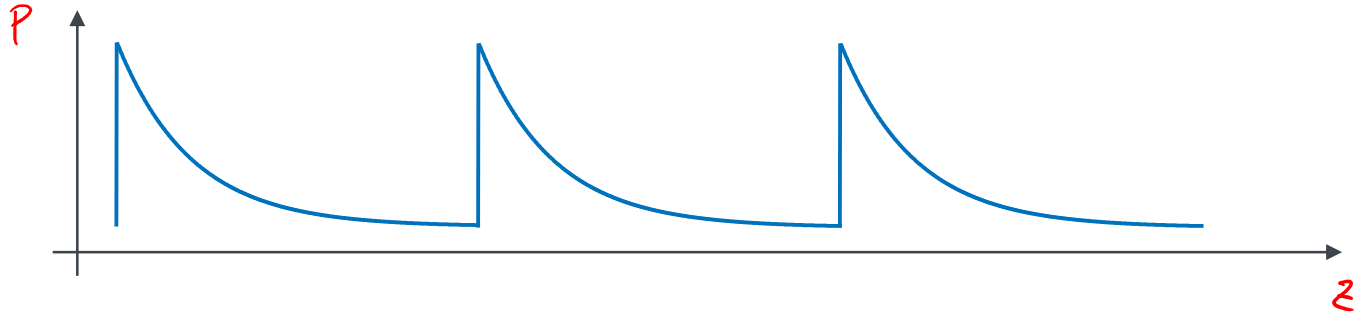


Prof. Dr.-Ing. Georg Rademacher

Power profile in amplified transmission link

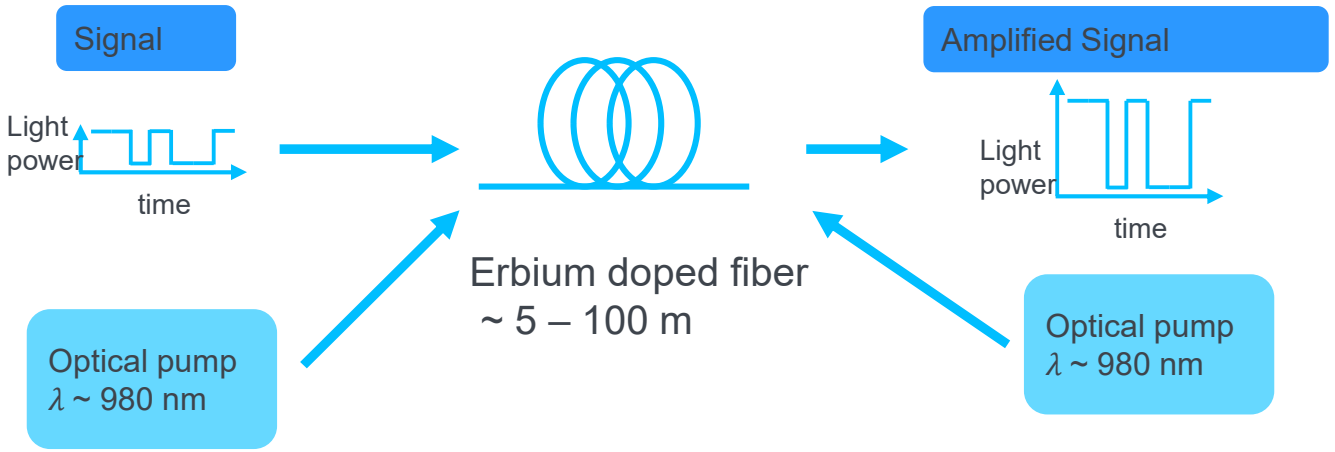
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Germany

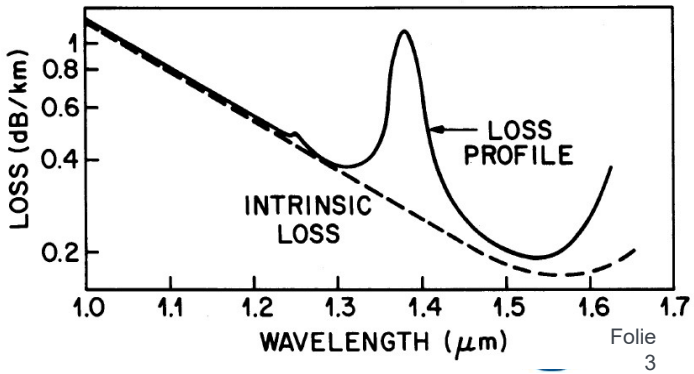




Erbium doped fiber amplifier (EDFA)

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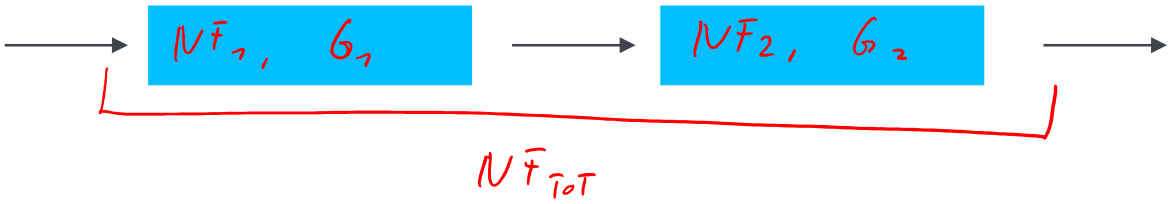


Noise in long distance transmission

University of Stuttgart
Germany

$$NF = 2 n_{sp} \left(1 - \frac{1}{G}\right) + \frac{1}{G} = \frac{1}{G}\left(1 + 2 n_{sp} (G - 1)\right) \approx 2 \cdot n_{sp} \quad (G >> 1)$$

$\Rightarrow$ minimum $NF = 3\,dB$



$$NF_{TOT} = NF_1 + \frac{NF_2 - 1}{G_1}$$

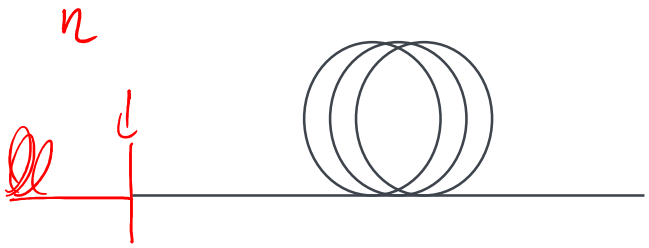
Noise in long distance transmission

University of Stuttgart
Germany

$$NF_1 = \frac{1}{G_1} = \frac{1}{\eta}$$

$$NF_2 = 2n_{sp}\left(1 - \frac{1}{G_2}\right) + \frac{1}{G_2}$$

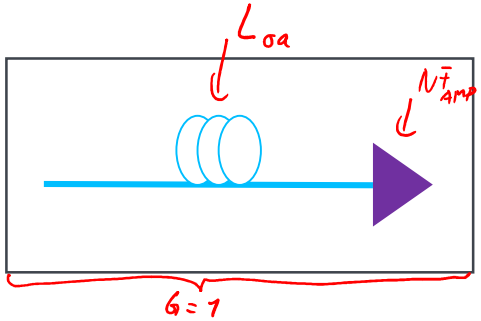
$$NF_{TOT} = \frac{1}{\eta} + \frac{NF_2 - 1}{\eta} = \frac{NF_2}{\eta}$$



$\rightarrow \triangleright NF_2$

Noise in long distance transmission

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Germany



$$N\bar{F}_{seg} = G + G(N\bar{F}_{amp} - 1) = G \cdot N\bar{F}_{AMP}$$

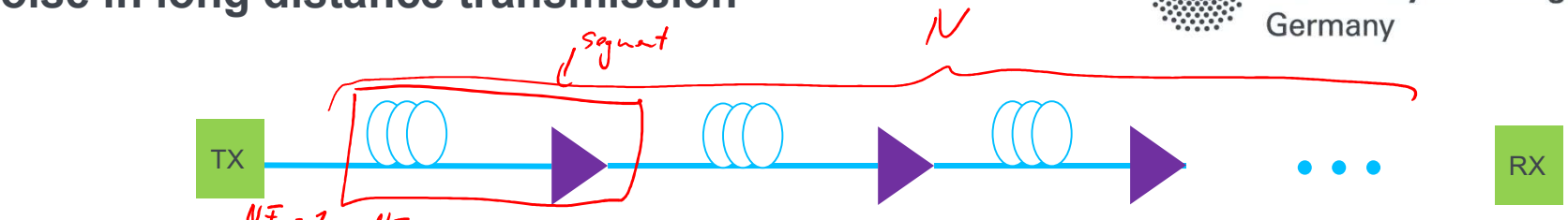
$$N\bar{F}_{seg} = 1 + 2n_{sp}(G-1)$$

Loss / attenuation : $\exp(-2\alpha L_{oa})$ $\rightarrow$ $G = \exp(2 \cdot \alpha \cdot L_{oa})$

α: Field-attenuation

Noise in long distance transmission

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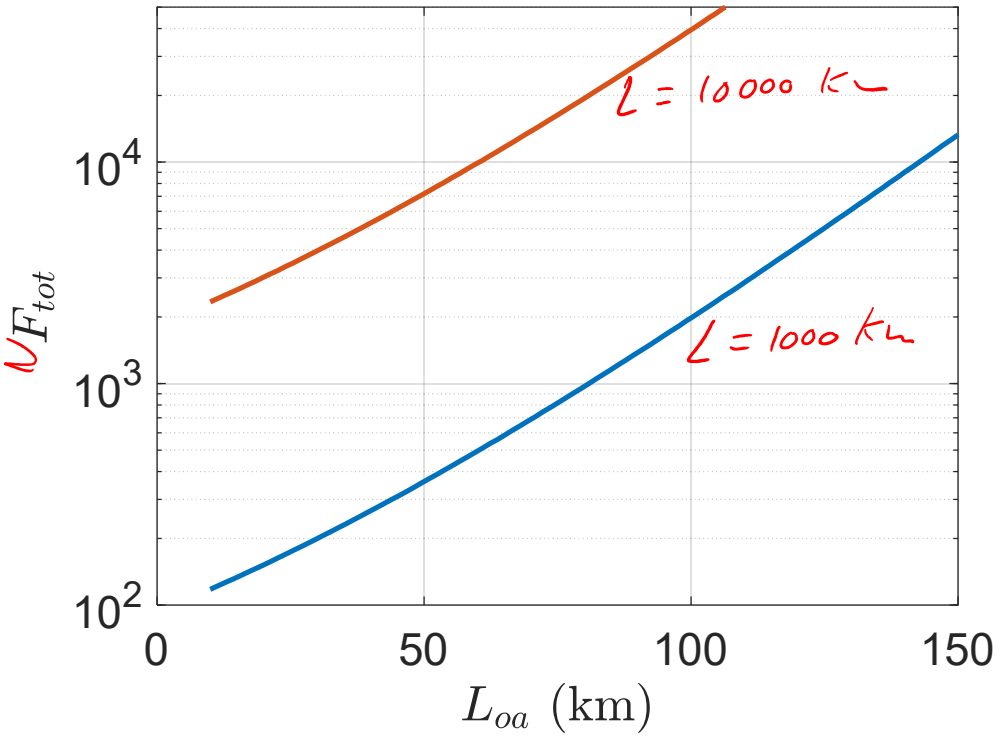
$$N\bar{F}_{TOT} = \bar{F}_1 + \frac{N\bar{F}_2 - 1}{G_1} + \frac{N\bar{F}_3 - 1}{G_1 \cdot G_2} + \dots$$

$$N\bar{F}_{TOT} = 1 + N(N\bar{F}_{seg} - 1) = 1 + 2N n_{sp}(G - 1)$$

Link distance: $L \rightarrow N = \frac{L}{L_{oa}}$, $G = \exp(2 \cdot \alpha \cdot L_{oa})$

$$\rightarrow N\bar{F}_{TOT} = 1 + 2 n_{sp} \frac{L}{L_{oa}} (\exp(2 \alpha L_{oa}) - 1)$$

Noise in long distance transmission



Optical Signal to Noise Ratio (OSNR)

University of Stuttgart
Germany

$$OSNR = \frac{P_0}{NF \cdot h \cdot f \cdot \Delta f}$$

$$\Delta f = 12{,}5\,GHz \hat{=} 0{,}1\,nm \quad (\sim 1550\,nm)$$

$$OSNR_{dB} = 10 \cdot \log_{10}(OSNR)$$

$$OSNR_{dB} = 58\,dB + P_0|_{dBm} - NF|_{dB}$$

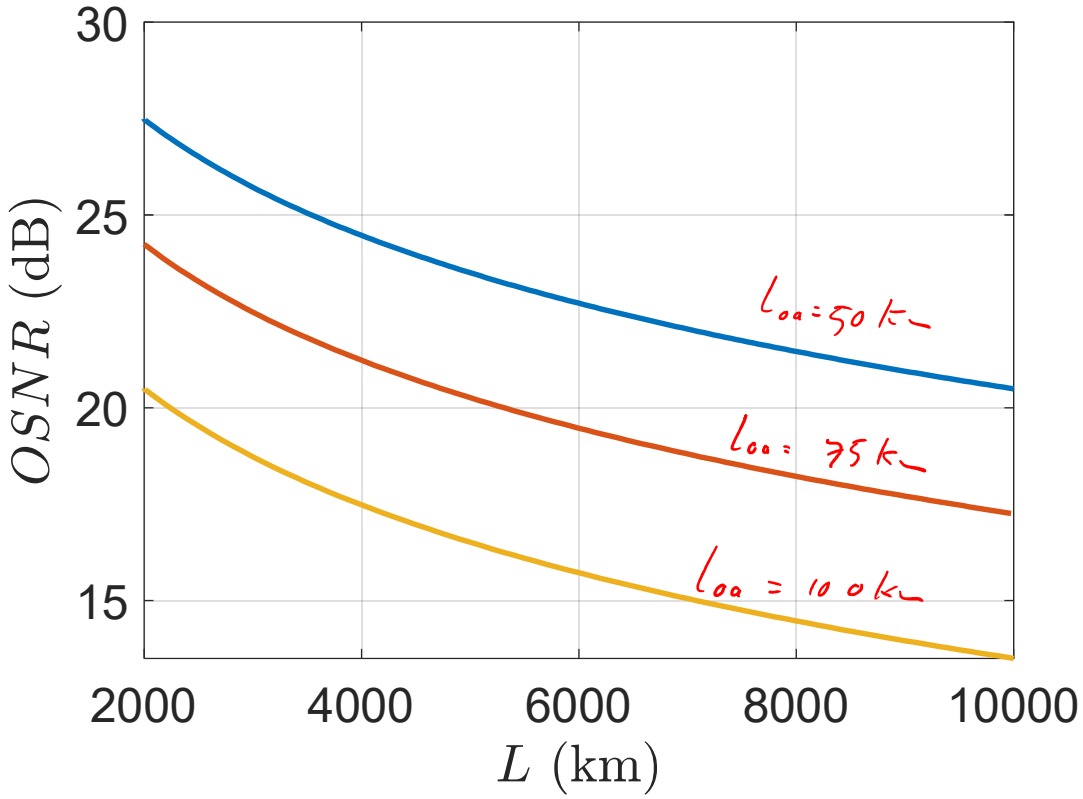
OSNR in long distance transmission

University of Stuttgart
Germany

$$OSNR|_{dB} = 58\,dB + P_0|_{dBm} - NF_{AMP} - \alpha \cdot L_{oa}|_{dB} - 10 \cdot \log(N)|_{dB}$$

OSNR in long distance transmission

University of Stuttgart
Germany



$\alpha = 0{,}2$ dB/km

$NF = 4{,}5$ dB



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Concept Test

Concept test

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Germany

What is the OSNR for P_0 = 2 mW, F_{Amp} = 4 dB, α = 0.2 dB/km, L_{oa} = 80 km, and N = 8?

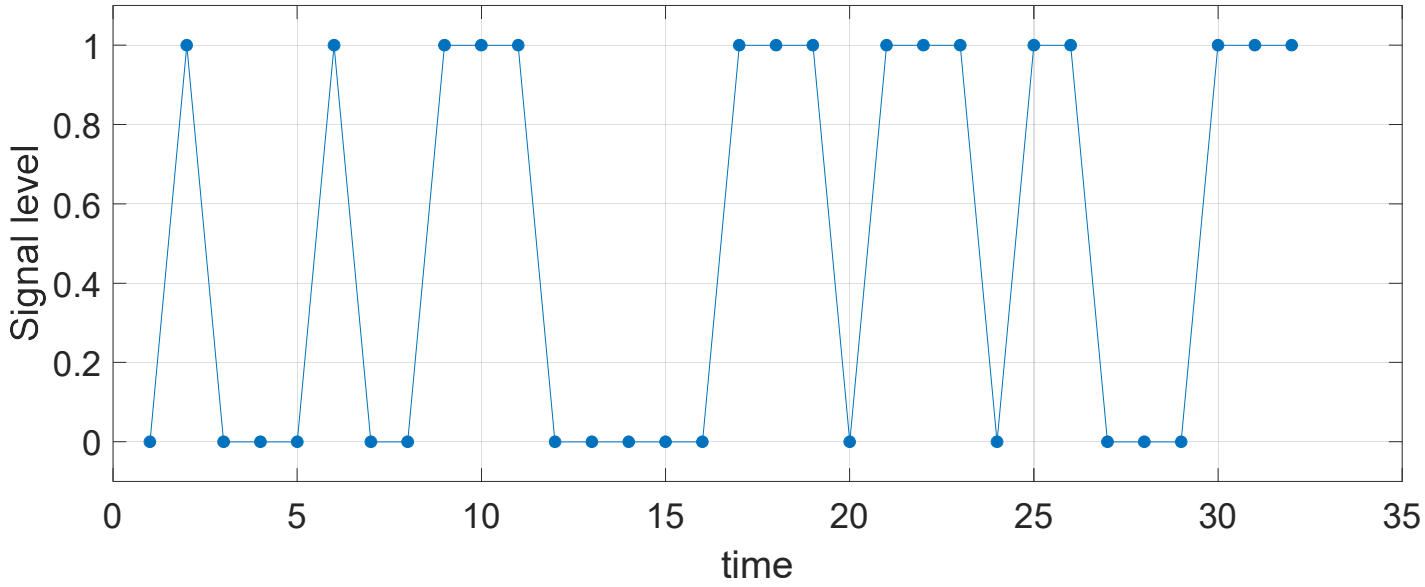
- **A** 44 dB
- **B** 30 dB
- **C** 50 dB

58 dB + 3 dBm - 4 dB - 16 dB -

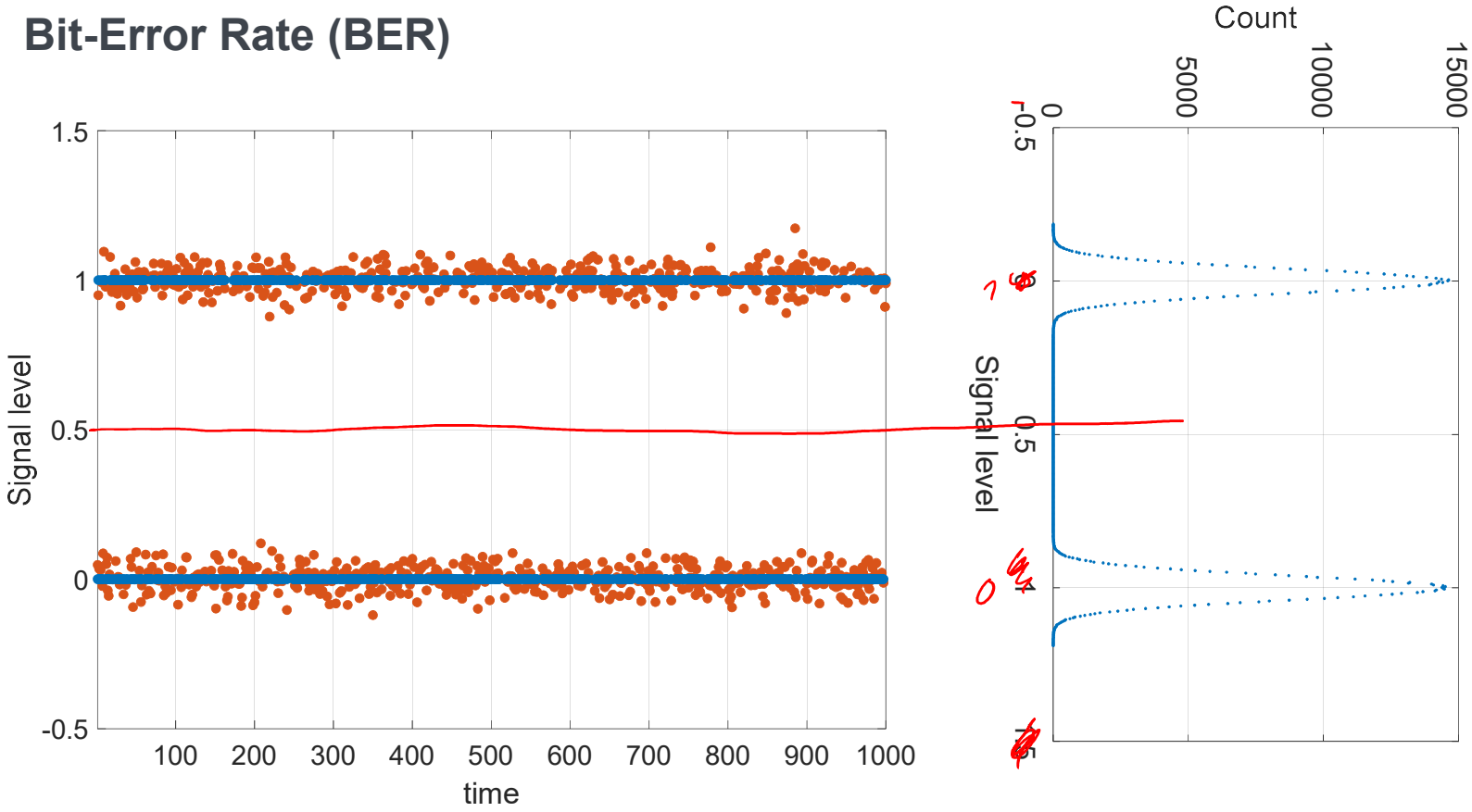
https://ilias3.uni-stuttgart.de/vote/KPS5

Bit-Error Rate (BER)

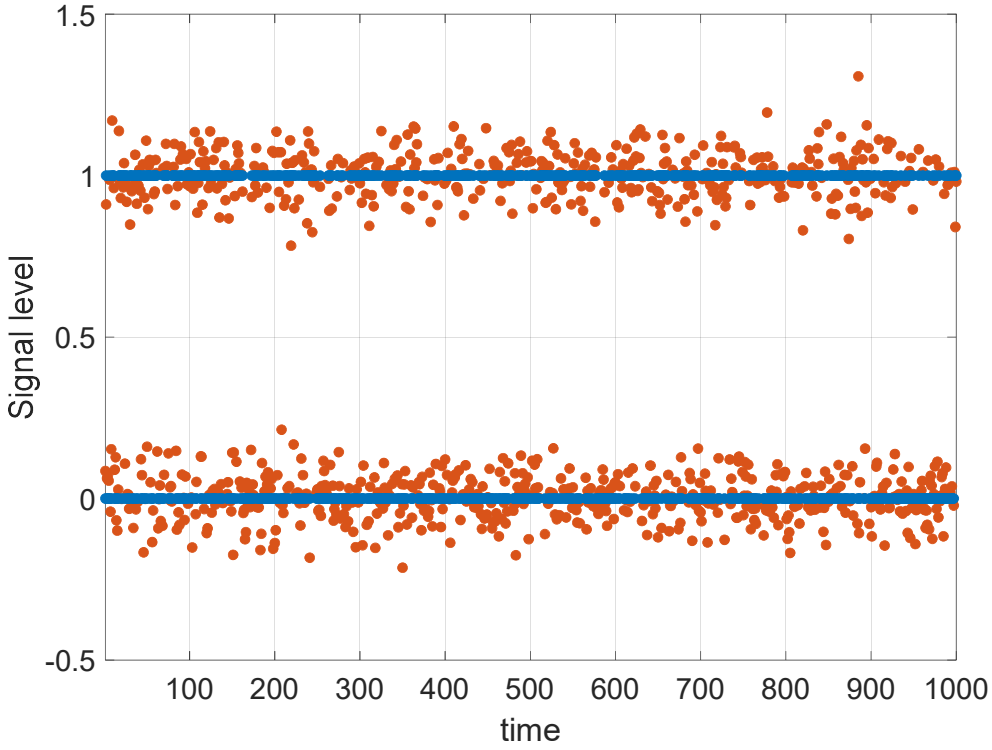
University of Stuttgart
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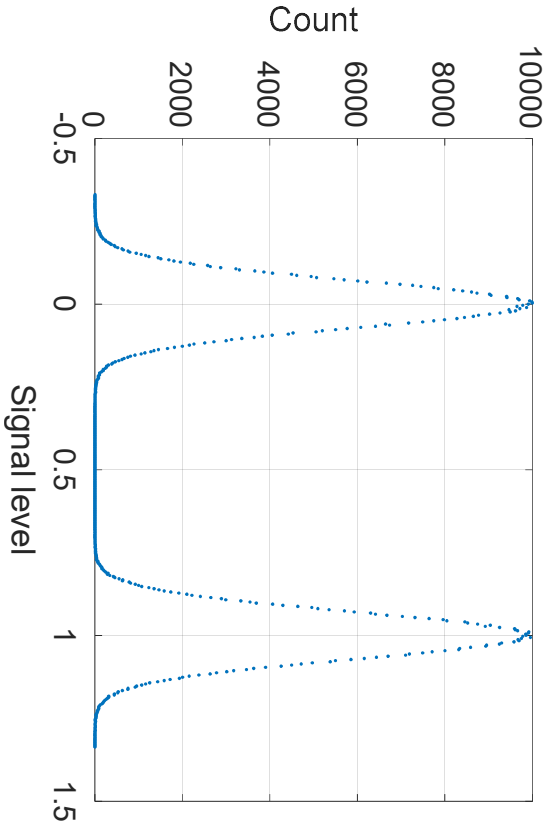


Bit-Error Rate (BER)

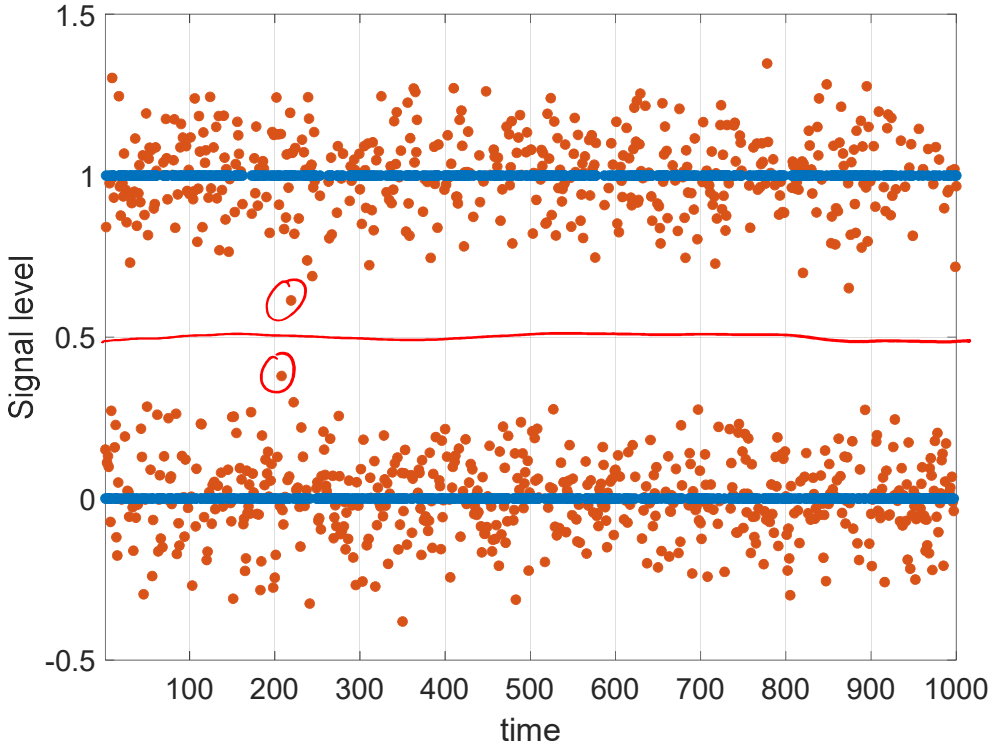


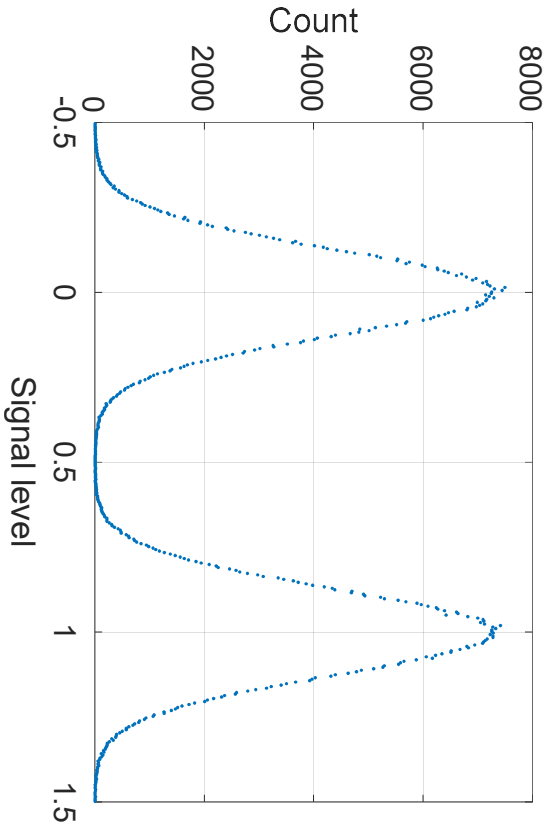
Bit-Error Rate (BER)



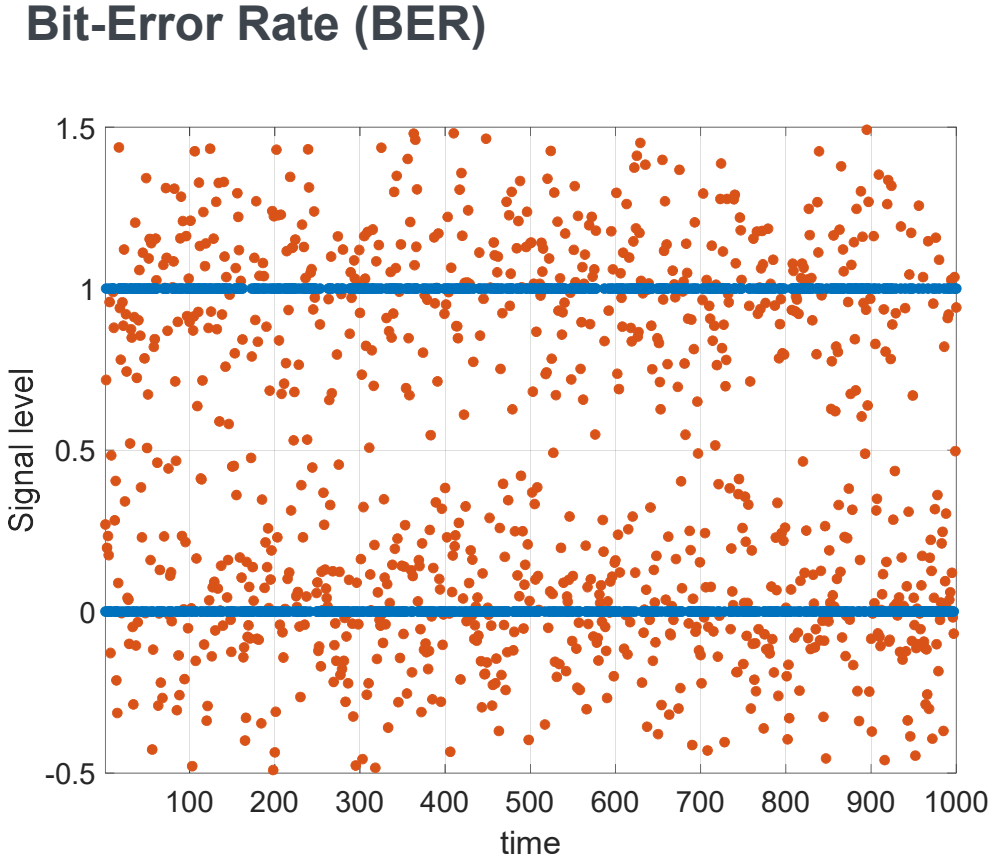


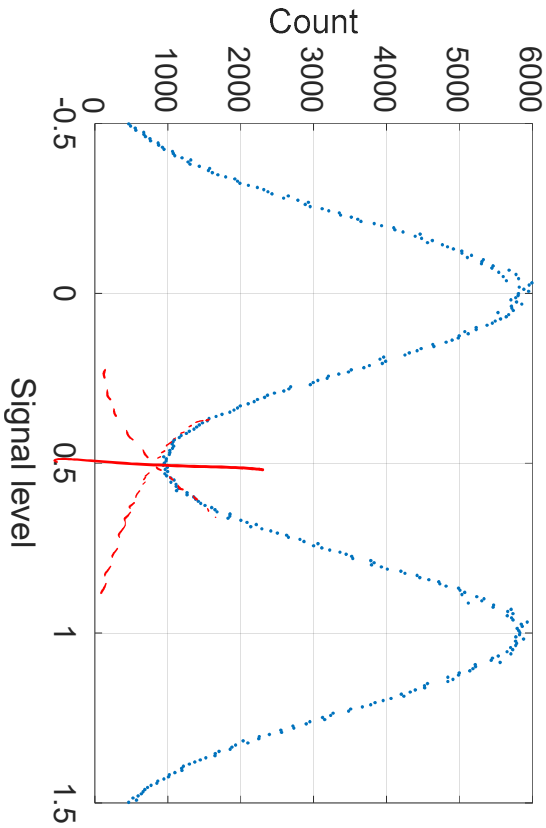
Bit-Error Rate (BER)





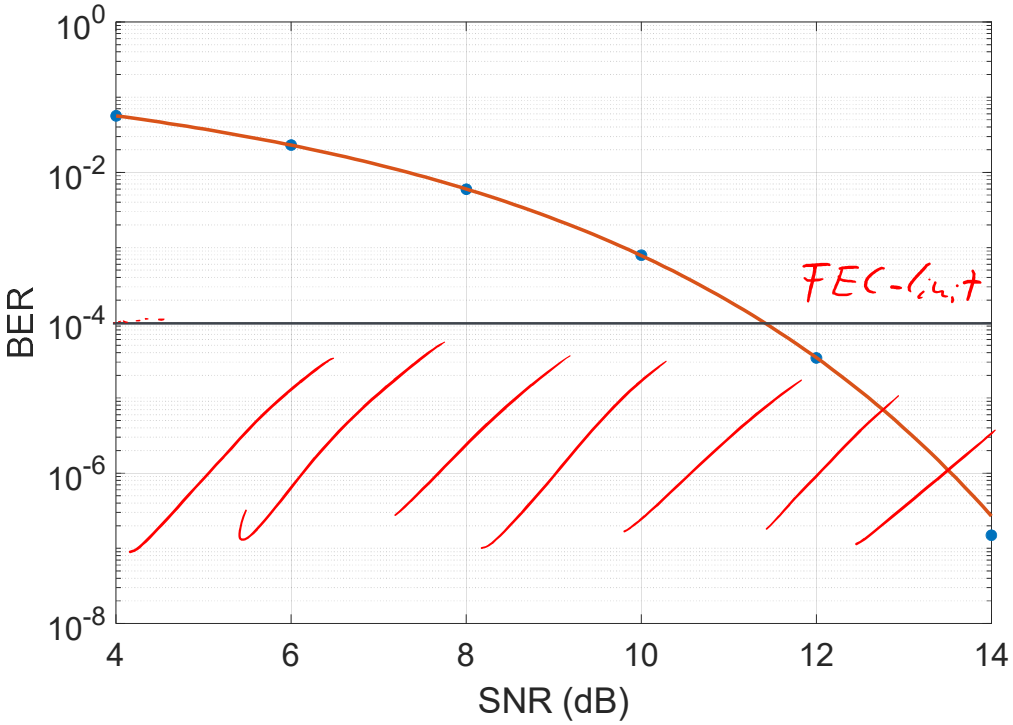
Bit-Error Rate (BER)





BER vs SNR for OOK

University of Stuttgart
Germany



$$BER = \frac{\#\ \text{wrong bits}}{\text{total}\ \#\ \text{bits}}$$

$$BER_{TH} = 0{,}5\ \text{erfc}\left(\sqrt{\frac{SNR}{2}}\right)$$

$$\text{erfc} = 1 - \text{erf}$$

$$\text{erf} = \frac{2}{\sqrt{\pi}} \int_0^x \exp(-t^2)\,dt$$

Forward Error Correction (FEC)

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Germany

(3,1) repitition code

Payload bit	Msg	received Msg.	interpreted Msg
0	000	000	0
		001	0
		010	0
		100	0
1	111	111	1
		110	1
		101	1
		011	1

Forward Error Correction (FEC)



University of Stuttgart
Germany

Information-Theoretic Tools for Optical Communications Engineers

(Invited Tutorial)

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TABLE I
HARD-DECISION FEC THRESHOLDS OF CERTAIN CODES RECOMMENDED FOR OPTICAL COMMUNICATIONS

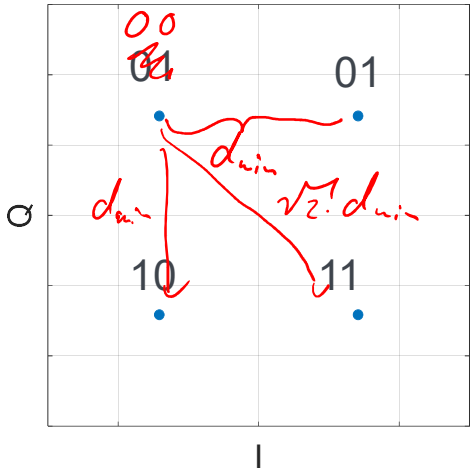
OH	Code type	Source	post-FEC BER	p
2.7%	Reed–Solomon (RS) (528,514) "KR4"	[26][a]	10^{-15}	$2.18 \cdot 10^{-5}$
5.8%	RS(544,514) "KP4"	[26][a]	10^{-15}	$2.26 \cdot 10^{-4}$
6.7%	RS(255,239)	[27][a]	10^{-15}	$8.27 \cdot 10^{-5}$
6.7%	Bose–Chaudhuri–Hocquenghem (BCH)+BCH	[28, App. I.3]	10^{-15}	$3.15 \cdot 10^{-3}$

Coherent Signals

4 QAM / QPSK

$\log_2(M) = 2$ bit/symbol

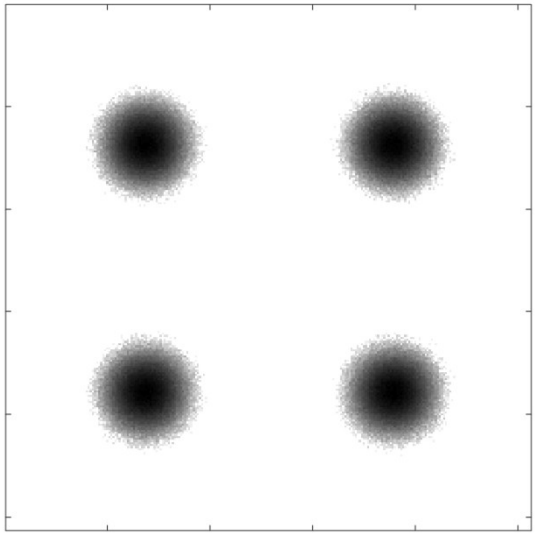
Gray - Coding



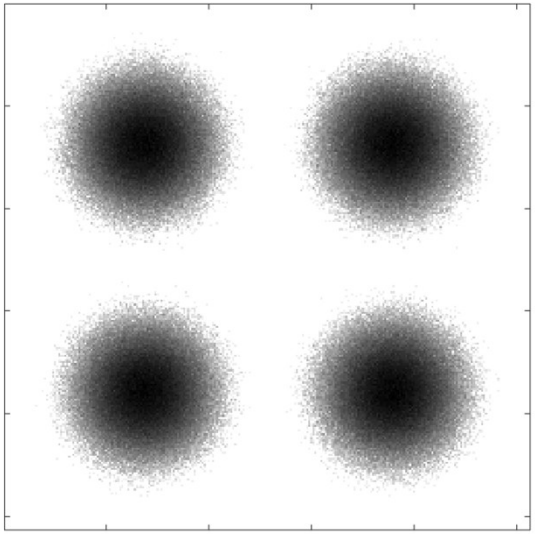
Coherent Signals

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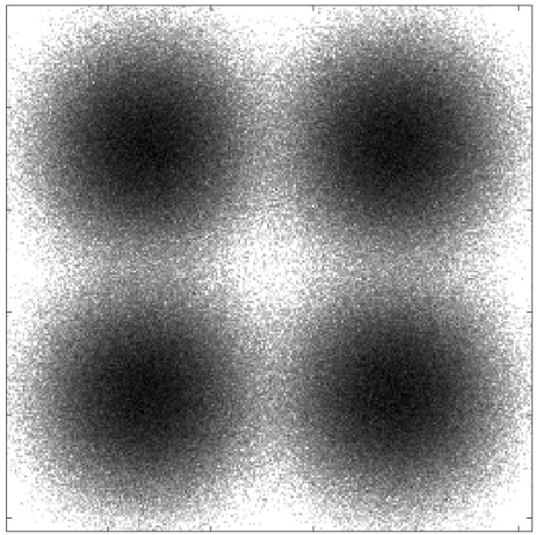
20 dB SNR



15 dB SNR



10 dB SNR



Coherent Signals

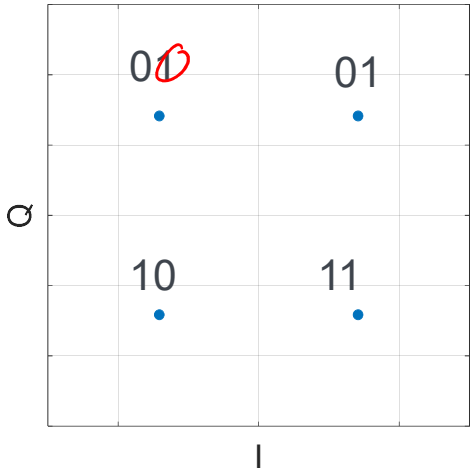
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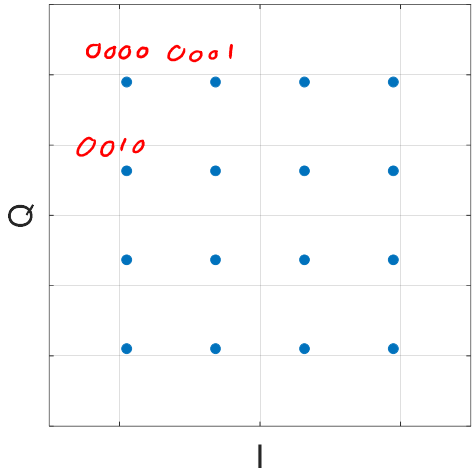
16 QAM

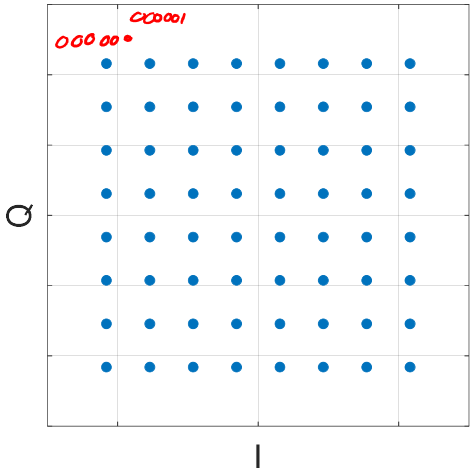
$\log_2(16) = 4\ bit/sy-b$

64 QAM

$\log_2(64) = 6\ bit/sy-b$







Data-rate $25\ GBd \cdot n\ \frac{bit}{sy-b} \cdot \frac{1}{(1 + OH_{FEC})}$



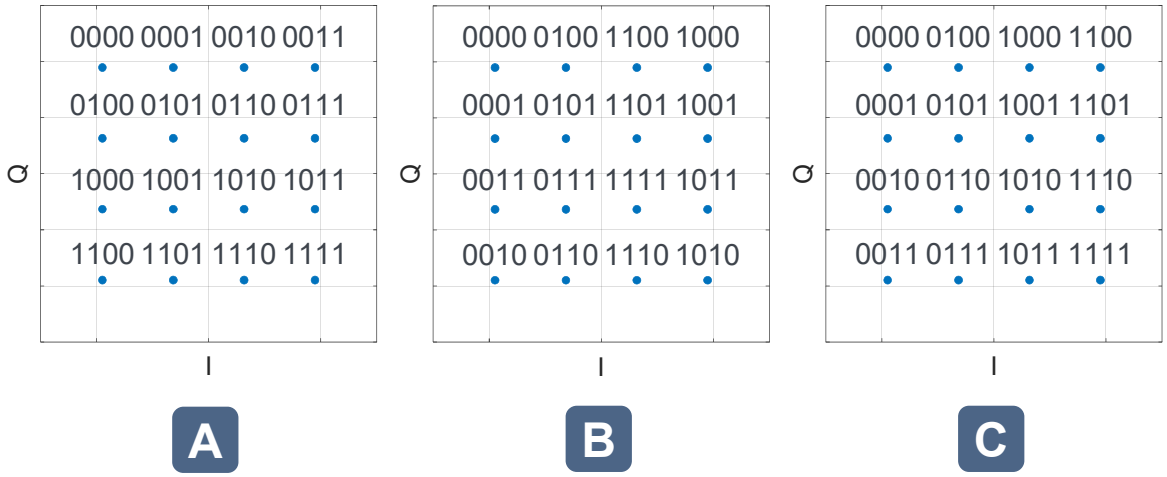
University of Stuttgart
Germany

Concept Test

Concept test

University of Stuttgart
Germany

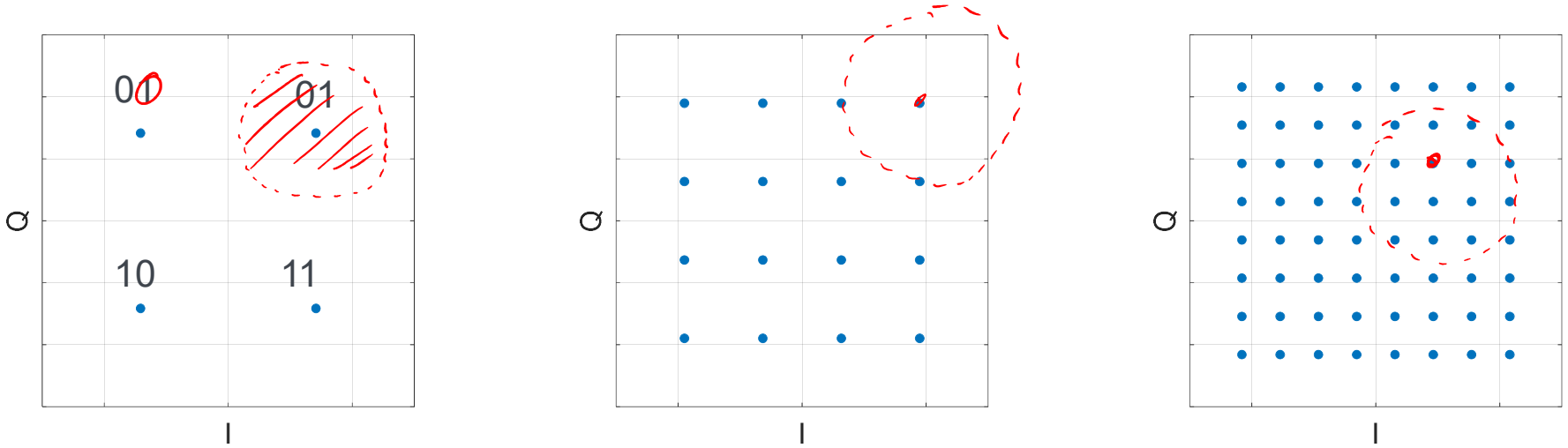
Which of the shown constellation diagrams is Gray encoded?



https://ilias3.uni-stuttgart.de/vote/PLKM

Coherent Signals

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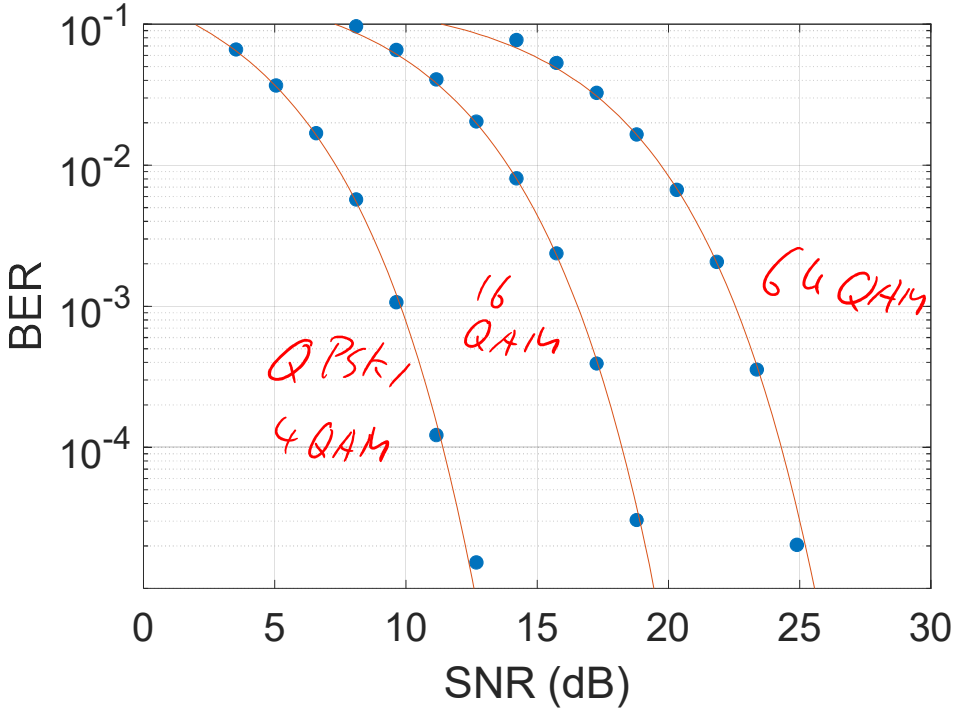
BER vs. SNR

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Germany

$$P_M = \left(1 - \frac{1}{\sqrt{M}}\right) \cdot \operatorname{erfc}\left(\sqrt{\frac{3}{2(M-1)} \cdot SNR}\right)$$

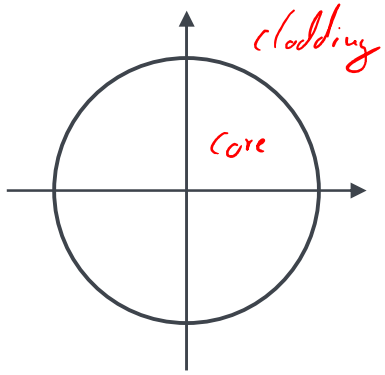
$$P_S = 1 - (1 - P_M)^2$$

$$P_e = \frac{P_S}{\log_2(M)} = BER$$



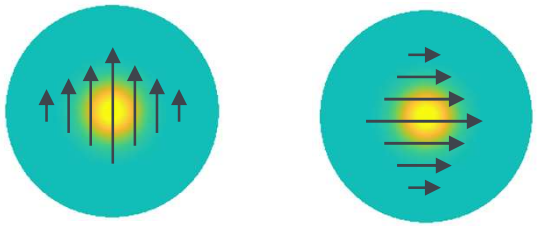
Polarization multiplex (Pol Mux)

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Germany

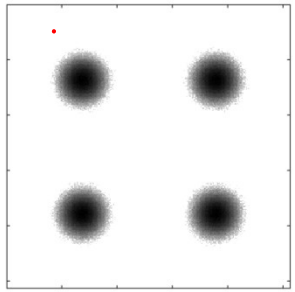


$E_y = 0$, $E_x = \psi(r,\varphi) \exp(-j\beta z)$

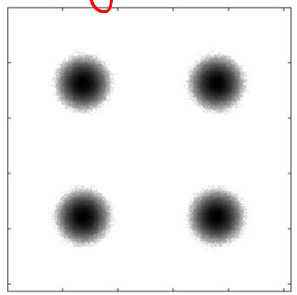
$E_y = \psi(r,\varphi) \cdot \exp(-j\beta z)$, $E_x = 0$



x- Pol

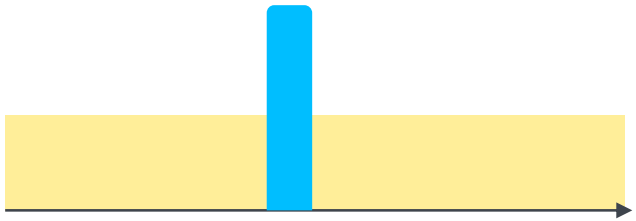


y- Pol



$25\,\text{GBd} \cdot n \frac{\text{bit}}{\text{Symb}} \cdot \frac{1}{1+OH_{FEC}} \cdot 2\,\text{pol}$

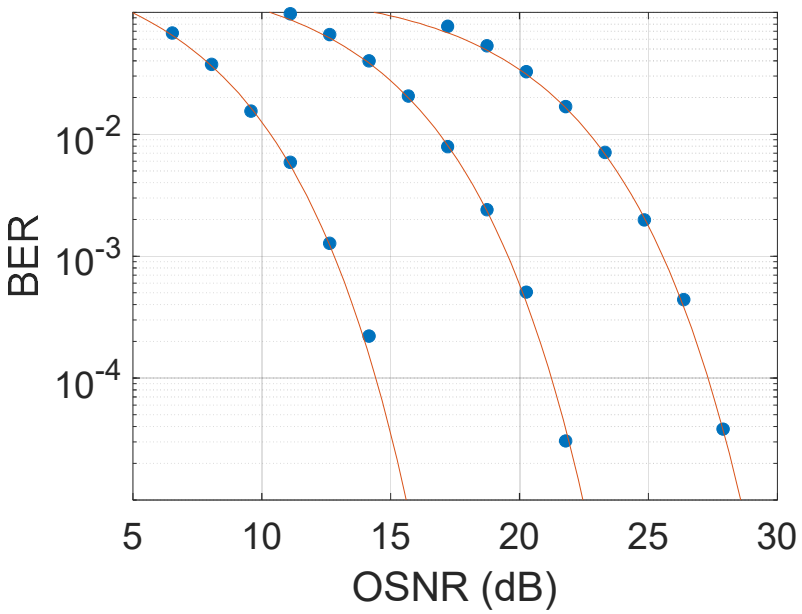
SNR vs. OSNR

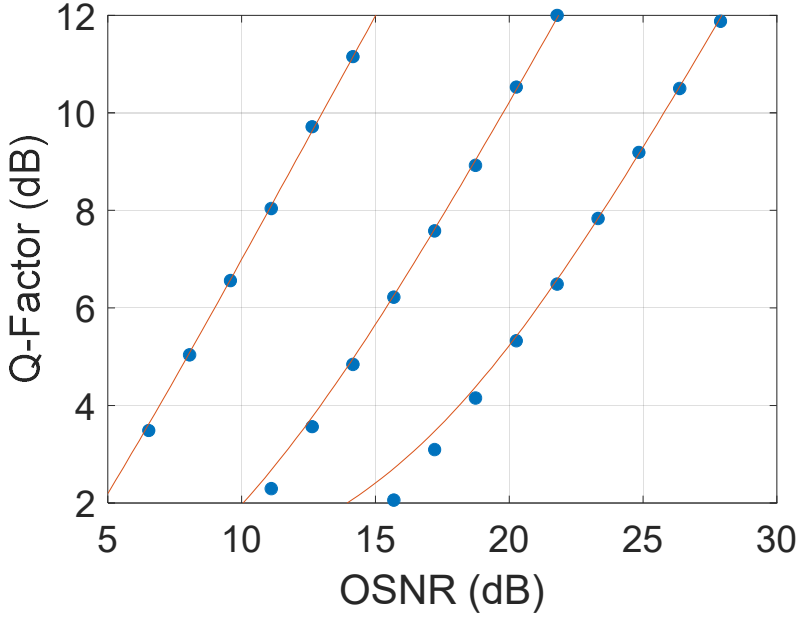




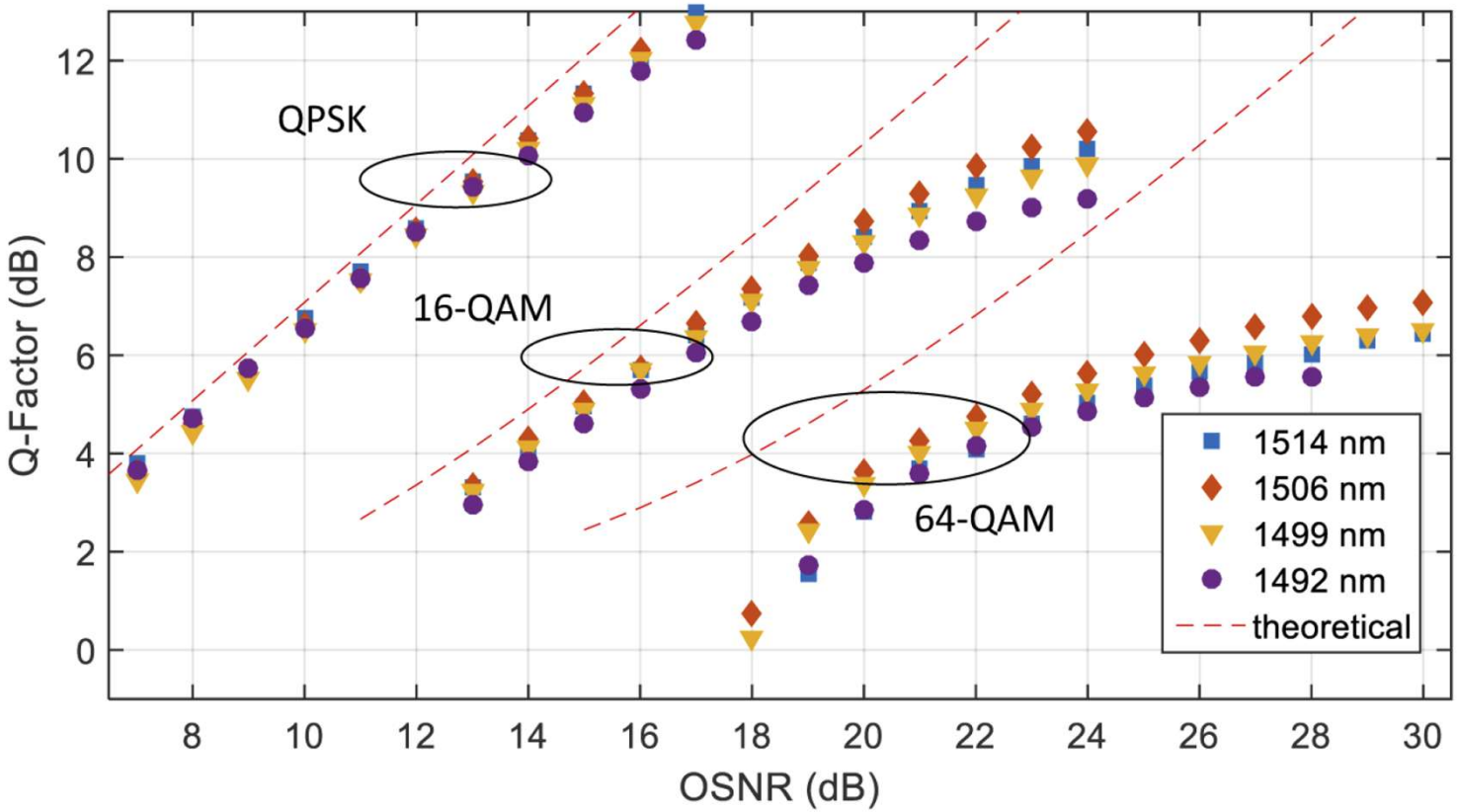
Coherent Signals

University of Stuttgart
Germany



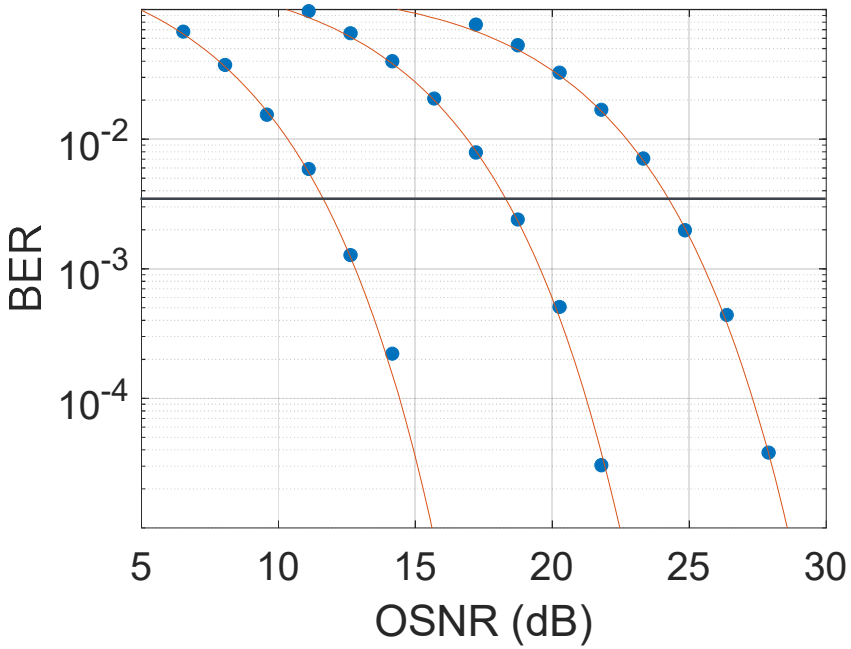


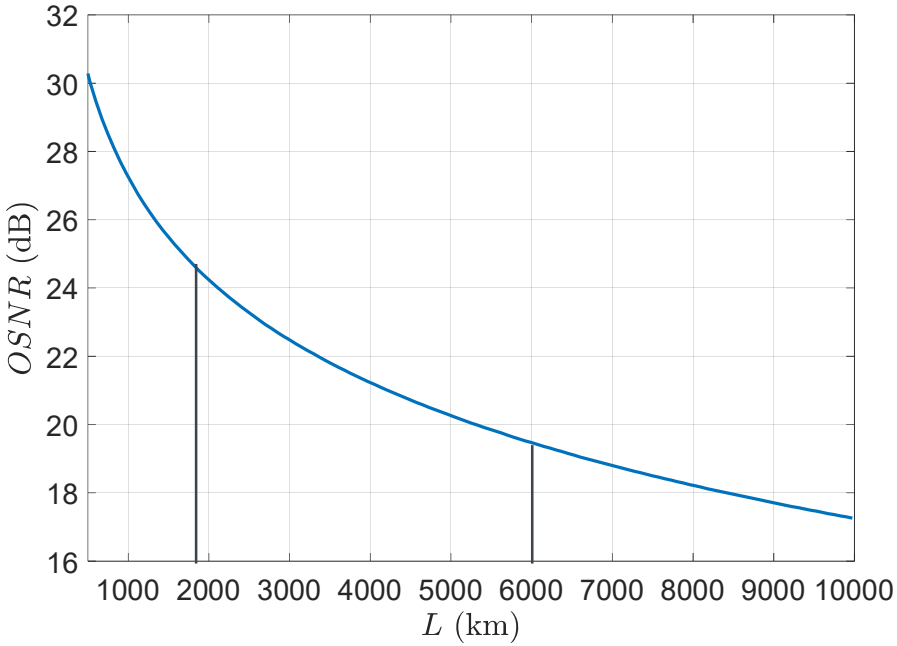
Measured performance



Long distance transmission

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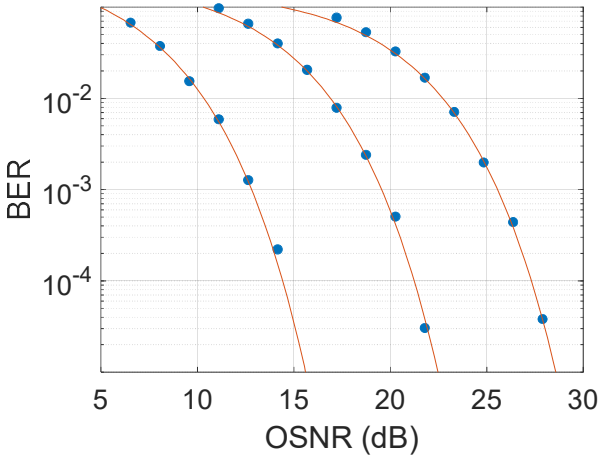
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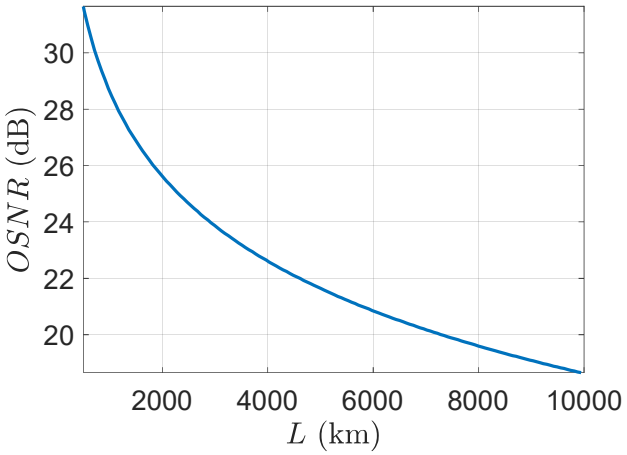
Concept Test

Concept test

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What is the maximum data rate for an FEC limit of $3 \cdot 10^{-4}$ with an overhead of 10% for L = 4500 km at 25 Gbd?

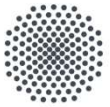




A 45.45 Gbit/s **B** 90.91 Gbit/s **C** 181.82 Gbit/s

https://ilias3.uni-stuttgart.de/vote/DC90

Pulse propagation in SMF



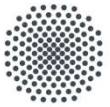
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Pulse propagation in SMF



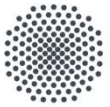
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Pulse propagation in SMF



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Pulse propagation in SMF



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Pulse propagation in SMF

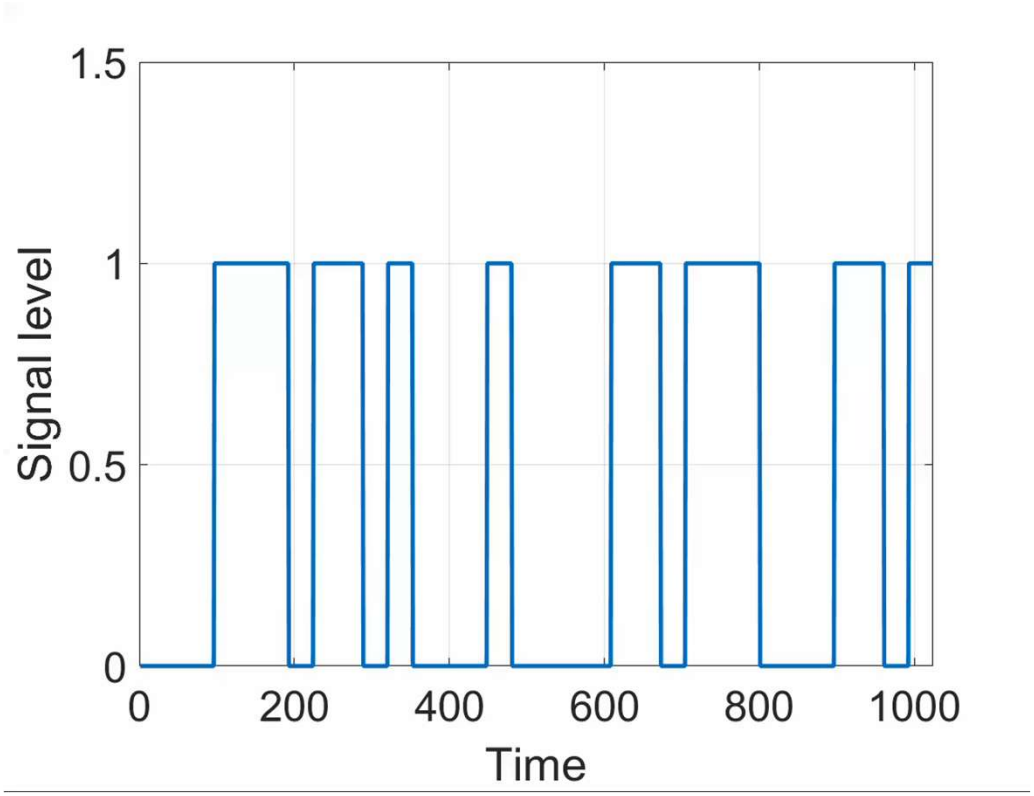


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Pulse propagation in SMF



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1.5
1
0.5
0
Signal level
0
200
400
600
800
1000
Time

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Institute of electrical and optical communications

Thank you!



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